Four Models

and a Common Fourth-Order Closure

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Physical fields, engineering tolerances,
and numerical verification

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Physical quantities and the purpose of the calculation

This manuscript develops four variational field models and the engineering calculations needed to assign tolerances to their numerical outputs. Every calculation is expressed through physical fields, units, material or coupling coefficients, boundary and initial data, discretized balance equations, and calibrated measurements. A common finite perturbation expansion is the starting point. Dimensional analysis, conditioning, resolution, and independent numerical verification determine what can actually be reported.

The four selected formulations are first-order vacuum gravity, fixed-Hamiltonian Schrödinger evolution, incompressible Navier–Stokes with response fields, and bare QED on a fixed background. Their action expansions terminate by fourth order in a declared field perturbation. The text derives their equations and examines their different physical response and numerical requirements. Beam force and energy calculations, Jacobian-free Newton–Krylov solution, multigrid error reduction, Fourier phase errors, and fluid and electromagnetic verification cases supply concrete tolerance calculations.

The scalar example retains exact integer arithmetic and an independent rational interval certificate. The instrument discussion then follows the signal from a calculated or measured quantity through calibration, storage, quantization, and a comparison between indicated values. The closing beam calculation restores physical stiffness, drag, work, heat, and temperature before defining an entropy reading.

A verified numerical calculation establishes accuracy for its stated model and inputs. An experimental comparison additionally tests the physical model and its calibration. The author's closing physical thesis is retained with the extra material and regularity estimate that it would require. The companion geometric specification supplies the separate enclosure and recovery construction.

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Chapter 1

Physical variables, action, and engineering response

A tolerance belongs to a quantity, a unit, an operating range, and a measurement procedure. A deflection tolerance of one micrometre and a relative pressure residual of one part per million cannot be compared by looking at their printed exponents. The same requirement applies to a fourth derivative: the independent variable, the dependent quantity, and its normalization have to be stated before a zero or a small number has an engineering interpretation.

The common calculation in this manuscript concerns a controlled change of independent fields in a variational formulation. Its result is an exact termination of a perturbation expansion. A separate calculation concerns the spatial bending of a slender beam. A third concerns the residual left by a numerical solution. We develop all three using field components, forces, energy, matrices, and calibrated output quantities. Their different units and boundary conditions stay visible throughout.

1.1 Field components and calibrated perturbations

For a fluid calculation the unknowns may be velocity components in metres per second and pressure in pascals. For an elastic calculation they may be displacement in metres and rotations in radians. For a wave calculation they include complex amplitudes, with real and imaginary components stored separately. Gravity and electrodynamics add metric, connection, potential, and spinor components with their chosen coordinate and unit conventions. Chapter 2 specifies the actual independent variables in each selected action.

After a mesh or a mode expansion is chosen, collect the independent coefficients in a column $a = (a_1, \dots, a_N)^T$. Before differentiating, fix the component scales $a_{0j} > 0$ and the reference coefficients a_j^{ref}. The dimensionless solver coordinates are

$$z_j = \frac{a_j - a_j^{\mathrm{ref}}}{a_{0j}}, \qquad a = a^{\mathrm{ref}} + D_a z, \qquad D_a = \operatorname{diag}(a_{01}, \dots, a_{0N}).$$

The scales are fixed for the perturbation being tested. A velocity increment and a pressure increment therefore enter the same numerical column only after their separate dimensional scales have been applied. Changing the unit from metres to millimetres changes D_a and the printed coordinates, while preserving the physical field.

Choose a trial field Z and a perturbation field H with the same units and boundary restrictions. Here Z denotes the complete list of independent fields; its numerical coefficients are the entries of a. The straight perturbation is $Z + sH$, with dimensionless amplitude s. The word straight means that the independent coefficients change linearly in s. It does not require the perturbed fields to solve the equations of motion. Such a variation is called off shell in physics. A perturbation of a metric must stay nondegenerate over the interval used. A perturbation of a clamped displacement has zero displacement and zero slope at its fixed endpoints.

Calibration here fixes the fields, component scales, sign conventions, boundary data, and output conversion used for the comparison. It does not insert a new term in a field equation. In the fermion calculation, the anticommutation convention is also specified explicitly; a calibration cannot turn anticommuting coefficients into ordinary commuting real numbers.

For a dynamical action S, choose an action scale $S_0 > 0$ with units of joule seconds and define

$$\phi(s) = \frac{S[Z + sH]}{S_0}. \tag{1.1}$$

Then ϕ and s are dimensionless, so their fourth derivative is dimensionless. The scale S_0 may be a declared characteristic energy times a characteristic time; a quantum calculation may use $\hbar$. A static elastic energy instead has units of joules and requires its own energy scale. A static energy and a dynamical action have different units even when both are minimized or made stationary by variation.

1.2 Why the perturbation expansion terminates

Consider a real-coefficient discrete action consisting of constant, linear, quadratic, and cubic terms,

$$S(a) = S_c + \sum_i b_i a_i + \frac{1}{2}\sum_{ij} K_{ij} a_i a_j + \frac{1}{6}\sum_{ijk} C_{ijk} a_i a_j a_k.$$

The coefficients include the quadrature weights, differential matrices, material constants, and fixed backgrounds. In this displayed real example they may be symmetrized without changing S. Substitution of $a + s\delta a$ gives

$$S(a + s\delta a) = c_0 + c_1 s + c_2 s^2 + c_3 s^3.$$

Each original term has at most three field factors. Selecting either the unperturbed value or the perturbation in each factor cannot produce a fourth power of s. A fixed spatial derivative is linear and preserves this count. Integration with fixed weights also preserves it.

Theorem 1.1 (Termination of the action perturbation expansion)**.** *If the selected action contains at most three independent-field factors in every term, with fixed coefficients and linear differential operators, then $D^4 S = 0$ and $\phi^{(4)} = 0$. Its variational equations, expressed with fixed coefficient scales, contain at most quadratic field terms, so their third perturbation derivative vanishes: $D^3 R = 0$.*

Proof. The substitution above gives the complete amplitude expansion. Differentiating it four times gives zero. The same count holds when four independent perturbation amplitudes are introduced: there is no term containing their product. Differentiating the action once to form the variational equations removes one field factor. Multiplication by fixed scaling matrices does not add one. The resulting residual is at most quadratic, and three further amplitude derivatives give zero. □

For the real coefficient model, the residual is explicitly

$$R_i(a) = b_i + \sum_j K_{ij} a_j + \frac{1}{2} \sum_{jk} C_{ijk} a_j a_k, \qquad J_{ij}(a) = \frac{\partial R_i}{\partial a_j}.$$

The matrix J is the tangent stiffness or Newton Jacobian. In a continuum calculation the corresponding residual is the coefficient of an arbitrary virtual displacement or field perturbation after integration by parts. Its boundary terms must be included or eliminated by the stated boundary conditions. In QED the expansion is performed with the fermion factors in their retained order. The zero fourth variation is then an ordered-coefficient identity in the declared fermion formulation, rather than a norm estimate for quantum states.

The condition describes a particular formulation. Eliminating an independent connection, solving a constraint through nonlinear coordinates, introducing a state-dependent normalization, or changing material coefficients with the unknown field can introduce further powers or denominators. For instance, a displacement-dependent tangent stiffness may still be a useful engineering model, but it requires its own perturbation count. A common zero fourth variation does not assert that four models have the same stiffness, characteristic speed, or measurement sensitivity.

Example 1.2 (Changing the perturbation coordinate)**.** The dimensionless response $\phi(s) = s^3$ has zero fourth derivative. If the controller instead reports $s = t^2$, the reported response is t^6 and its fourth derivative with respect to t is $360t^2$. The independent perturbation amplitude is part of the test. Physical time is another variable: a harmonic displacement $q(t) = q_0 \cos \omega t$ satisfies $q^{(4)} = \omega^4 q$, even when its mechanical action is quadratic.

1.3 Force residuals, tangent response, and least squares

Many discretized field balances have linear and quadratic terms. In calibrated coefficient coordinates write

$$R(z) = Lz + B(z, z) - f, \tag{1.2}$$

where L is a fixed matrix or linear differential operator and $B_i(z, w) = \sum_{jk} B_{ijk} z_j w_k$ is a bilinear interaction. The fluid convective term is an example, with $B(u, w) = (u \cdot \nabla)w$ before pressure and other terms are assembled. A direct expansion gives

$$DR(z)h = Lh + B(h, z) + B(z, h), \tag{1.3}$$

$$D^2R(z)[h, k] = B(h, k) + B(k, h), \qquad D^3R = 0. \tag{1.4}$$

Both interaction terms are part of the usual Newton linearization. For fluid velocity they are $(h \cdot \nabla)u$ and $(u \cdot \nabla)h$. Retaining the second perturbation derivative is useful for an error estimate or a higher-order update; it does not supply a convective contribution missing from the exact Jacobian.

Residual components must be scaled before their squares are added. A force imbalance in newtons, a divergence defect in inverse seconds, and a boundary displacement error in metres are separate channels. After division by their declared reference values, use fixed positive weights w_i and define

$$\Phi(z) = \frac{1}{2} \sum_i w_i R_i(z)^2, \qquad \|R\|_W^2 = \sum_i w_i R_i^2.$$

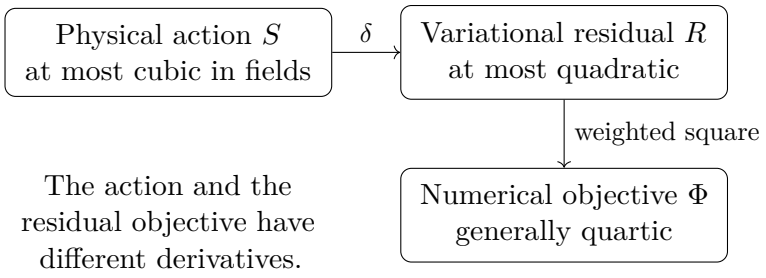


Figure 1.1: Action, field balance, and numerical objective. The field units, residual scales, and fixed weights determine the meaning of each derivative.

The weights can include cell-volume fractions or sensor weighting. For the unit-weight convention used in the displayed equation below, $R(z + sh) = R(z) + sJ(z)h + s^2B(h, h)$ implies

$$\frac{\mathrm{d}^4}{\mathrm{d}s^4}\Phi(z + sh) = 12\,\|B(h, h)\|_2^2\,. \tag{1.5}$$

With fixed weights the right side is $12\sum_i w_i B_i(h, h)^2$. Thus a squared residual can be quartic while the physical action is cubic. The distinction affects a numerical minimization: stationarity of Φ gives $J^T W R = 0$, whereas a solved balance requires $R = 0$. The elementary residual $R(z) = z^2 + 1$ has a least-squares stationary point at $z = 0$ and no real zero. A solver must report the balance defect as well as a change in its objective.

1.4 Beam curvature, bending energy, and virtual work

Let x measure distance along a straight beam of length L, and let $q(x)$ be transverse displacement. In the Euler–Bernoulli model the slope is small, material response is linear elastic, and transverse shear deformation is neglected. Young's modulus E has units $\mathrm{N\,m^{-2}}$; the cross-section's second moment of area I_b has units $\mathrm{m^4}$. Write $EI = EI_b$ for the constant flexural rigidity, with units $\mathrm{N\,m^2}$. The approximate curvature is q'', with units $\mathrm{m^{-1}}$.

The bending energy and total potential for a distributed transverse load $f(x)$ are

$$U[q] = \frac{EI}{2}\int_0^L (q'')^2\,\mathrm{d}x, \qquad \Pi[q] = U[q] - \int_0^L fq\,\mathrm{d}x.$$

Here f has units $\mathrm{N\,m^{-1}}$ and both terms in Π are joules. Choose the energy-conjugate bending moment $M_b = EIq''$; its sign fixes the following moment and shear convention. Twice integrating the virtual work gives

$$\delta\Pi = [M_b\delta q' - M_b'\delta q]_0^L + \int_0^L (M_b'' - f)\delta q\,\mathrm{d}x.$$

For clamped ends, $\delta q = \delta q' = 0$ at both supports. The interior balance is $(EIq'')'' = f$, or $EIq^{(4)} = f$ for constant rigidity. At an unloaded free end the independent virtual displacement and rotation instead give $M_b' = 0$ and $M_b = 0$. These force, moment, constitutive, and boundary relations are the engineering content of the variational calculation [22].

With zero distributed load and constant rigidity, $q^{(4)} = 0$ between loads. It is a spatial balance, not

the action derivative in (1.1). The normalized bending functional is

$$E[q] = \frac{U[q]}{EI} = \frac{1}{2}\int_0^L (q'')^2 \,\mathrm{d}x. \tag{1.6}$$

This normalization removes the stiffness; unless position and displacement are also nondimensionalized, $E[q]$ has units of inverse metres. It is distinct from Young's modulus, which always occurs in EI in the beam formulas. A fully dimensionless energy uses $\xi = x/L$, $q = d\widehat{q}$, and scale $U_0 = EId^2/L^3$.

Proposition 1.3 (Exact clamped-beam energy gap)**.** *Let q_* be the unloaded cubic satisfying four fixed endpoint displacements and slopes. For a perturbation v with $v = v' = 0$ at both ends and finite integrated squared curvature,*

$$U[q_* + v] - U[q_*] = \frac{EI}{2}\int_0^L (v'')^2 \,\mathrm{d}x, \qquad E[q_* + v] - E[q_*] = \frac{1}{2}\left\|v''\right\|_{L^2}^2.$$

Consequently q_* is the unique minimum-energy deflection for those data.

Proof. Expanding the bending energy leaves the cross term $EI\int q_*''v''$. Its boundary terms vanish, and its interior term is $EI\int q_*^{(4)}v = 0$. The remaining squared-curvature term is nonnegative. If it vanishes, $v'' = 0$, so v is a straight displacement and rotation. The homogeneous clamp conditions force both to zero. □

1.5 A clamped displacement with dimensional reactions

Impose $q(0) = q'(0) = q'(L) = 0$ and $q(L) = d$. Solving the four coefficient equations gives

$$q_*(x) = d(3\xi^2 - 2\xi^3), \qquad \xi = x/L, \qquad U[q_*] = \frac{6EId^2}{L^3}.$$

The curvature is $(6d/L^2)(1 - 2\xi)$ and the third derivative is $-12d/L^3$. The support moments have magnitude $6EI|d|/L^2$ and the transverse reaction magnitude is $12EI|d|/L^3$. Differentiating the energy with respect to the imposed displacement gives that same reaction: $\partial U/\partial d = 12EId/L^3$. Quasistatic work from zero displacement to d is the integral of this linear reaction, $6EId^2/L^3$, which equals the stored bending energy.

For a declared example with $EI = 1\,\mathrm{N\,m^2}$, $L = 1\,\mathrm{m}$, and $d = 10^{-3}\,\mathrm{m}$, the stored energy is $6 \times 10^{-6}\,\mathrm{J}$, the reaction magnitude is $0.012\,\mathrm{N}$, and each end-moment magnitude is $0.006\,\mathrm{N\,m}$. The maximum slope is $1.5d/L = 0.0015$. This is a calculation in the stated linear beam model; the stiffness and dimensions are design inputs rather than measured material data. In dimensionless coordinates $d = L = 1$, the normalized energy is the previous scalar value $E[q_*] = 6$.

An applied distributed load changes the solution order. For example, a simply supported beam with constant load f_0 has

$$q(x) = \frac{f_0}{24EI}(L^3x - 2Lx^3 + x^4), \qquad q(L/2) = \frac{5f_0L^4}{384EI}.$$

Direct differentiation gives $EIq^{(4)} = f_0$; both endpoint displacements and endpoint bending moments vanish. The quartic term is the required load response. A variable stiffness likewise requires $(EI(x)q'')'' = f$. Neither case permits replacing that balance by $q^{(4)} = 0$ without retaining its load or material terms.

1.6 Cubic beam elements and spline reactions

A two-node Euler–Bernoulli element uses the physical degrees of freedom $d_e = (q_0, \theta_0, q_L, \theta_L)^T$. Its displacement is

$$q = N_1 q_0 + L N_2 \theta_0 + N_3 q_L + L N_4 \theta_L,$$

where

$$N_1 = 1 - 3\xi^2 + 2\xi^3, \quad N_2 = \xi - 2\xi^2 + \xi^3,$$
$$N_3 = 3\xi^2 - 2\xi^3, \qquad N_4 = -\xi^2 + \xi^3.$$

The factors of L are essential because rotations and displacements have different units. Inserting the interpolation into U gives $U_e = \frac{1}{2} d_e^T K_e d_e$, with

$$K_e = \frac{EI}{L^3} \begin{pmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{pmatrix}.$$

The force entries of $K_e d_e$ have units of newtons and its moment entries have units of newton metres. A free element has two zero-energy motions, translation and rigid rotation. Suitable fixed supports remove these motions and make the assembled elastic solve uniquely determined. The same element exactly represents the unloaded clamped cubic. Exact representation of this case is a useful verification test, while a loaded quartic still needs mesh refinement or an enriched interpolation.

For several interpolation knots, the deflection and slope are retained at adjoining elements. Minimization with free interior slopes gives continuous bending moment when there is no applied concentrated couple. For uniform stiffness this is continuity of q''. A jump in EIq''' represents a concentrated transverse reaction. Consequently a curve can be cubic on every open segment while its global fourth spatial derivative includes point-load terms. Natural spline endpoint curvature is zero when the endpoint slopes are free and no endpoint moment is applied. These are the classical spline minimum-energy conditions [5, 4]; their engineering interpretation is moment and shear balance at the joins.

1.7 Converting a beam force residual into deflection error

For one explicit tolerance calculation, suppose the true and computed beam deflections have identical fixed clamps and enough smoothness for integration by parts. Put $e = q_h - q_*$ and let $r = EIe^{(4)}$ be the remaining distributed force imbalance. A force residual is useful only if this equation, the material coefficient, and its homogeneous boundary conditions are the ones being tested. Multiplying by e gives

$$EI \left\| e'' \right\|_{L^2}^2 = \int_0^L re \, \mathrm{d}x.$$

Two one-dimensional Poincare estimates, or a sine-series estimate with these homogeneous endpoints, give $\|e\|_{L^2} \le (L/\pi)^2 \|e''\|_{L^2}$. Hence

$$\left\| e'' \right\|_{L^2} \le \frac{L^2}{EI\pi^2} \left\| r \right\|_{L^2}, \qquad \left\| e \right\|_{L^2} \le \frac{L^4}{EI\pi^4} \left\| r \right\|_{L^2}.$$

A point displacement follows from $e(x) = \int_0^x (x - s)e''(s)\,\mathrm{d}s$ and Cauchy–Schwarz. Writing $r_{\mathrm{rms}} = (L^{-1}\int_0^L r^2\,\mathrm{d}x)^{1/2}$ yields the conservative engineering bound

$$\max_x |e(x)| \le \frac{L^4}{\sqrt{3}\,EI\pi^2}\, r_{\mathrm{rms}}.$$

Its units are metres. In the preceding one-metre, unit-rigidity example, $r_{\mathrm{rms}} \le 10^{-3}\,\mathrm{N\,m^{-1}}$ gives a displacement bound of approximately $5.85 \times 10^{-5}\,\mathrm{m}$. A one-micrometre specification would require a proportionally smaller residual or a sharper, verified response estimate.

This calculation assumes that concentrated reactions and boundary errors have already been accounted for. An element join with an unbalanced shear can have zero open-element fourth derivative and a nonzero point force. Ignoring that force would invalidate the load residual used above. The engineering tolerance therefore records the interior balance, interface forces and moments, support data, material uncertainty, and the displacement output separately. The four action formulations will require the same discipline with their own physical fields and response operators.

Chapter 2

Four field models, their actions, and physical scales

The common fourth-variation closure has a concrete realization in four field actions. The fields represent spacetime geometry, a quantum wavefunction, an incompressible viscous fluid, and electromagnetic interaction with a charged spinor. Their action integrals describe stationarity with respect to independent changes in these fields. Each realization depends on a choice of independent variables. Retaining a connection makes the gravitational action cubic; retaining response fields gives a cubic stationary formulation of viscous flow. The fixed-Hamiltonian Schrödinger action is quadratic, while the bare Maxwell–Dirac action has a cubic interaction. These statements concern the actions before solving constraints, eliminating fields, or passing to an effective theory.

We derive the stationary equations as well as the degree count. This distinguishes three operations: adjoining auxiliary fields, projecting a stationary system onto its physical fields, and eliminating those auxiliaries inside an action. Projection can preserve the solution set without preserving polynomial degree, a norm, or a variational minimum. Throughout, derivatives of fields count linearly: replacing Z by $Z + sH$ also replaces ∂Z by $\partial Z + s\partial H$.

Presentation	Independent fields	Action degree
Vacuum gravity	Densitized inverse metric, torsion-free connection	≤ 3
Schrödinger	Complex wavefunction and its conjugate variation	≤ 2
Navier–Stokes	Velocity, pressure, two response fields	≤ 3
Bare QED	Gauge potential, spinor, independent adjoint spinor	≤ 3

The boundary sectors below are part of these presentations. Compactly supported variations suffice for the local field equations. Whenever an endpoint is free, its boundary term is retained explicitly. A new boundary action changes the variational problem and must receive its own degree count.

The device's two reading directions name gravity and fluid response *covariant*, and the wavefunction and QED presentations *contravariant*. Those names organize this comparison. The concrete actions below determine the mathematical content; raising or lowering an index does not turn one stationary system into another.

2.1 First-order vacuum gravity

Work on a four-dimensional spacetime region admitting a smooth Lorentzian metric. In a coordinate chart write $g = \det(g_{\mu\nu}) < 0$ and introduce the symmetric contravariant tensor density

$$h^{\mu\nu} = \sqrt{-g}\, g^{\mu\nu}.$$

The independent fields are h and a torsion-free affine connection $\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$. Nondegeneracy and Lorentzian signature define the admissible open sector for h; affine variations remain in it for sufficiently small parameter intervals. There is no assumption that the independently varied connection is already metric compatible.

With a fixed nonzero normalization suppressed, take

$$S_{\mathrm{GR}}[h,\Gamma] = \int h^{\mu\nu} R_{\mu\nu}(\Gamma)\, \mathrm{d}^4x, \tag{2.1}$$

where our curvature convention is

$$R_{\mu\nu}(\Gamma) = \partial_\lambda \Gamma^{\lambda}_{\mu\nu} - \partial_\nu \Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\lambda}_{\lambda\rho}\Gamma^{\rho}_{\mu\nu} - \Gamma^{\lambda}_{\nu\rho}\Gamma^{\rho}_{\mu\lambda}.$$

For dimensions, use coordinates $x^0 = ct, x^1, x^2, x^3$ measured in length, with dimensionless metric components. The dimensional action is $c^3 S_{\mathrm{GR}}/(16\pi G)$: Γ has units of inverse length, $R_{\mu\nu}$ of inverse length squared, and the integral has units of length squared. The prefactor therefore gives the action units of energy times time. Suppressing a nonzero constant changes neither stationarity nor the number of field factors. It does not mean that c or Newton's constant has disappeared from predictions expressed in SI units.

A reference length L gives a useful numerical form. Put $\widehat{x} = x/L$, $\widehat{\Gamma} = L\Gamma$, and $\widehat{R} = L^2 R$. Then

$$S^{\mathrm{SI}}_{\mathrm{GR}} = \frac{c^3 L^2}{16\pi G} \int h^{\mu\nu} \widehat{R}_{\mu\nu}(\widehat{\Gamma})\, \mathrm{d}^4\widehat{x}.$$

The dimensionless integral is the quantity a code may assemble. The length L must still accompany a quoted curvature tolerance: a componentwise residual of 10^{-8} in this integral corresponds to $10^{-8}/L^2$ in dimensional Ricci curvature. A tidal acceleration measurement requires a further conversion using the selected observer's frame and the separation of its test masses. In an orthonormal frame, curvature components $R_{\hat{0}\hat{i}\hat{0}\hat{j}}$ enter geodesic deviation with an acceleration scale $c^2\ell/L^2$ for separation ℓ. A small coordinate component has a physical interpretation only after that frame and length scale have been supplied.

The density makes the coordinate integral invariant. Its two kinds of monomial are $h\,\partial\Gamma$ and $h\Gamma\Gamma$, of degrees two and three. This is the first-order polynomial mechanism used in Deser's derivation; the following compatibility calculation fixes the precise torsion-free sector used here [6].

There is also a coordinate-independent meaning to the affine variation. The difference C of two connections is a tensor, so $(h,\Gamma) + s(k,C)$ is a legitimate path of independent fields. Its cubic coefficient in the chosen chart is

$$\int k^{\mu\nu} \left(C^{\lambda}_{\lambda\rho} C^{\rho}_{\mu\nu} - C^{\lambda}_{\nu\rho} C^{\rho}_{\mu\lambda} \right) \mathrm{d}^4x.$$

No term of fourth degree in s occurs. The same argument with several independent variation parameters establishes vanishing of the full fourth multilinear variation, rather than only one

selected diagonal derivative. Nondegeneracy restricts the admissible interval of the path; it does not insert a denominator into this independent-field action.

Here s is a dimensionless perturbation amplitude, not proper time, coordinate distance, or an additional spacetime coordinate. For example, one may choose k and LC to have unit size in a declared component norm, so that $|s|$ measures the imposed change relative to those field scales. The gravitational action then records the first response, second response, and cubic interaction of that particular perturbation. Its fourth derivative with respect to this amplitude is zero even when the resulting gravitational field has arbitrarily complicated coordinate dependence. In particular, this assertion does not remove curvature or make the fourth spatial derivative of a metric component vanish.

For a symmetric variation $k^{\mu\nu} = \delta h^{\mu\nu}$, stationarity immediately gives

$$R_{(\mu\nu)}(\Gamma) = 0. \tag{2.2}$$

For the connection variation use

$$\delta R_{\mu\nu} = \nabla_\lambda \delta\Gamma^\lambda_{\mu\nu} - \nabla_\nu \delta\Gamma^\lambda_{\mu\lambda}.$$

Integration by parts with compact support is legitimate for the weight-one density h. Symmetrizing in the two lower indices of $\delta\Gamma$ yields

$$-\nabla_\lambda h^{\mu\nu} + \frac{1}{2}\delta^\nu_\lambda D^\mu + \frac{1}{2}\delta^\mu_\lambda D^\nu = 0, \qquad D^\mu = \nabla_\rho h^{\mu\rho}. \tag{2.3}$$

Contracting λ with ν gives $3D^\mu/2 = 0$ in four dimensions. Thus the connection equation is exactly $\nabla_\lambda h^{\mu\nu} = 0$.

To recover ordinary metric compatibility, put $Q_{\lambda\mu\nu} = \nabla_\lambda g_{\mu\nu}$ and $q_\lambda = g^{\mu\nu}Q_{\lambda\mu\nu}$. The density derivative gives

$$0 = \nabla_\lambda h^{\mu\nu} = \sqrt{-g}\left(-g^{\mu\alpha}g^{\nu\beta}Q_{\lambda\alpha\beta} + \frac{1}{2}q_\lambda g^{\mu\nu}\right).$$

Lowering the free indices gives $Q_{\lambda\mu\nu} = q_\lambda g_{\mu\nu}/2$. Its trace is $q_\lambda = 2q_\lambda$, so $q = Q = 0$. The unique torsion-free metric-compatible connection is therefore the Levi–Civita connection of g. Its Ricci tensor is symmetric, and (2.2) becomes $R_{\mu\nu}(g) = 0$. Taking the trace shows this is equivalent to $R_{\mu\nu} - g_{\mu\nu}R/2 = 0$ in this dimension.

Proposition 2.1 (Vacuum stationary projection)**.** *For smooth nondegenerate Lorentzian h and torsion-free Γ, the interior stationary equations of* (2.1) *are equivalent to the vacuum Einstein equations with $\Lambda = 0$, together with $\Gamma = \Gamma(g)$. In these independent variables, $D^4 S_{\mathrm{GR}} = 0$.*

Proof. The two variations and contractions above give the forward implication. Conversely, a vacuum metric and its Levi–Civita connection satisfy both Euler–Lagrange equations. The degree conclusion follows from the displayed monomials and Theorem 1.1. □

Eliminating Γ substitutes an expression containing g^{-1} and derivatives of g; the resulting metric action need not be polynomial in an affine metric variation. A cosmological volume term, matter fields, torsion, or an unrestricted connection changes the hypotheses and can change both the degree and the auxiliary equations. The proposition is a local stationary equivalence; it neither fixes coordinates nor supplies an evolution estimate for a chosen gravitational observable.

The boundary choice has an engineering role. Fixing the independent connection at a boundary removes the corresponding surface variation; compact support avoids that issue when deriving local equations. A finite evolution calculation must in addition prescribe compatible initial data, a coordinate condition, and boundary treatment for incoming information. The light cones follow from the recovered Lorentzian metric. The present action does not impose the Lorentz transformation as an extra scalar multiplier, and a choice of coordinates is not itself a source of stress-energy.

Vacuum is also an explicit physical assumption. This action contains no fluid stress, electromagnetic stress, cosmological term, or material constitutive law. The dimensionless compactness $GM/(Lc^2)$ is useful when a mass M appears through the exterior solution or boundary data, but it is not an independently varied matter density in the action written here. A weak-field calculation can expand about Minkowski data; a strong-field calculation must additionally monitor nondegeneracy and coordinate conditioning. Both may use the same cubic first-order action. Neither obtains a bound on a detector displacement merely by counting its field factors. Curvature, coordinate constraints, and the selected detector response enter that bound separately. A flat but nonconstant coordinate example in Chapter 5 makes this distinction directly testable.

2.2 Fixed-Hamiltonian Schrödinger evolution

Use complex wavefunctions with finite integrated probability, $\mathcal{H} = L^2(\Omega; \mathbb{C})$, and the overlap integral $\langle \psi, \phi \rangle = \int_\Omega \bar{\psi}\phi \, \mathrm{d}x$. Fix a time-independent self-adjoint Hamiltonian H, with the field regularity and boundary conditions specified by $D(H)$. For example,

$$H = -\frac{\hbar^2}{2m}\Delta + V(x), \qquad m > 0,$$

with real V and boundary conditions that make this realization self-adjoint. The domain is part of the model: a differential expression alone does not determine H. Use paths regular enough that $\dot{\psi}$ and $H\psi$ belong to $\mathcal{H}$ and the following integrations exist. The physical equation and its probability interpretation are the ones in Tong's Sections 1.2–1.2.2 [19].

In d spatial dimensions a probability-normalized wavefunction has units length$^{-d/2}$, because $\int_\Omega |\psi|^2 \, \mathrm{d}x = 1$. The expectation $\langle \psi, H\psi \rangle$ is an energy, and both terms of the action below integrate to energy times time. The mass m, Planck constant $\hbar$, potential V, and spatial boundary conditions are prescribed physical inputs. They do not become additional field variations merely because an implementation stores them alongside the wavefunction.

For the free-particle kinetic term, take a length L and define

$$t_* = \frac{mL^2}{\hbar}, \qquad E_* = \frac{\hbar^2}{mL^2}, \qquad \widehat{\psi} = L^{d/2}\psi, \quad \tau = t/t_*, \quad \widehat{x} = x/L.$$

The equation becomes $i\partial_\tau \widehat{\psi} = [-\Delta_{\widehat{x}}/2 + V/E_*]\widehat{\psi}$. Consequently a dimensionless phase tolerance has a definite laboratory time scale, and a potential error must be divided by E_* before it is combined with a normalized kinetic residual. A different reference length changes these reported dimensionless numbers without changing the dimensional physical prediction.

On $[0, T]$ define the real action

$$S_{\mathrm{Sch}}[\psi] = \int_0^T \left[\frac{i\hbar}{2}(\langle \psi, \dot{\psi} \rangle - \langle \dot{\psi}, \psi \rangle) - \langle \psi, H\psi \rangle \right] \mathrm{d}t. \tag{2.4}$$

Varying real and imaginary parts independently is equivalent to treating ψ and $\bar{\psi}$ as independent coordinates during the formal variation and then imposing complex conjugacy. For variations vanishing at both time endpoints, integration by parts gives

$$\delta S_{\mathrm{Sch}} = 2\,\mathrm{Re}\int_0^T \langle \delta\psi, i\hbar\dot{\psi} - H\psi\rangle\,\mathrm{d}t.$$

Hence stationarity gives the evolution and its adjoint equation,

$$i\hbar\partial_t\psi = H\psi, \qquad -i\hbar\partial_t\langle\psi| = \langle\psi|H. \tag{2.5}$$

In the real-potential differential example the second equation is $-i\hbar\partial_t\bar{\psi} = H\bar{\psi}$. Conversely these equations make the interior variation zero. The endpoint restriction is a device for deriving the equation, not a requirement that a physical initial-value problem prescribe both initial and final wavefunctions.

More explicitly, before imposing endpoint restrictions the boundary contribution is

$$\frac{i\hbar}{2}\big(\langle\psi,\delta\psi\rangle - \langle\delta\psi,\psi\rangle\big)\Big|_0^T.$$

Thus unrestricted endpoint variation would define a different stationary problem. For an unbounded Hamiltonian, require both ψ and $H\psi$ to have finite squared integrals, and use the dimensionally compatible size $\|\psi\| + \|H\psi\|/E_*$. Choose time dependence regular enough for the stated integration by parts and perturbations preserving those conditions. This supplies the regularity needed for a strong solution, rather than merely differentiating a formal differential expression. Spatial boundary conditions have already entered through $D(H)$ and are not additional free variations at this step.

All terms in (2.4) are quadratic in the independent complex coordinates. Thus $D^3 S_{\mathrm{Sch}} = 0$, and its Euler–Lagrange residual is linear. In particular its fourth variation vanishes with one order to spare. This affine degree statement is made off shell: a variation $\psi + s\eta$ need not stay on the unit sphere. Solving a normalization constraint first would instead introduce a nonlinear parametrization of that sphere.

For a strong solution, self-adjointness also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\psi\|^2 = 2\,\mathrm{Re}\langle\psi, -iH\psi/\hbar\rangle = 0.$$

The spectral calculus supplies the unitary initial-value solution $\psi(t) = e^{-itH/\hbar}\psi(0)$, with strong differentiability for initial data in $D(H)$. Unitarity bounds state perturbations in this probability-amplitude norm; a numerical Fourier truncation still has its own projection and time-stepping errors. Even one energy eigenstate has time dependence $e^{-iEt/\hbar}$, whose fourth time derivative generally does not vanish. The action closure therefore concerns the declared variation parameter, independently of the solution's physical oscillation.

For the real scalar potential used above, direct substitution of the Schrödinger equation gives the local probability balance

$$\partial_t|\psi|^2 + \nabla\cdot j = 0, \qquad j = \frac{\hbar}{m}\,\mathrm{Im}(\bar{\psi}\nabla\psi).$$

Periodicity cancels the integrated boundary flux. An impermeable quantum boundary must instead make its normal probability current vanish; Dirichlet data or homogeneous Neumann data are

standard examples for the free kinetic operator on a suitable domain. Absorbing boundary conditions change this balance and cannot be included in a claim of exact norm preservation without also accounting for the absorbed flux. In a Fourier implementation, using periodic modes is therefore a physical and numerical boundary decision, rather than an innocuous change of basis.

The readout must be named with equal care. For a bounded self-adjoint observable A and normalized states ψ, ϕ,

$$|\langle \psi, A\psi\rangle - \langle \phi, A\phi\rangle| \le 2\,\|A\|\,\|\psi - \phi\|\,.$$

To see this, insert $\langle \phi, A\psi\rangle$ between the two expectations and apply Cauchy–Schwarz to each difference. A detector probability for a spatial region is the expectation of its projection operator, so $\|A\| = 1$ supplies an immediate probability-error bound. Energy and momentum are generally unbounded operators. An L^2 state error alone does not bound their errors; the energy-domain or derivative control required by those observables must be included in the numerical contract. Norm preservation is consequently a valuable conservation test, but does not certify the phase of an interference pattern or the energy content of an unresolved high-frequency tail.

This distinction also identifies the source of stiffness. For a Fourier mode with wave number k, the kinetic angular frequency is $\omega_k = \hbar|k|^2/(2m)$. Refining the spatial mesh introduces faster oscillations and tighter phase requirements, even though the action is still quadratic. A self-adjoint spectral discretization preserves the free mode frequencies exactly; a finite-difference discretization changes them through its discrete Laplacian. A variable potential also couples Fourier modes, so projection and multiplication must be assessed with the actual cutoff and quadrature. The computed phase example in Section 5.3 separates these spatial choices from the error of the time integrator.

2.3 Incompressible Navier–Stokes with response fields

Take a periodic spatial domain $\Omega = \mathbb{R}^d/\mathbb{Z}^d$, $d = 2$ or 3, and a finite time interval. Fix $\nu > 0$, a force f, and a divergence-free initial velocity u_0. Density has been normalized to one, so p is pressure divided by that density. The physical equations are the incompressible equations stated by Fefferman [8]; the response-field construction below is an explicit stationary representation of them.

Before normalizing density, the dimensional momentum equation is

$$\rho(\partial_t u + (u\cdot\nabla)u) = \nabla\cdot\sigma + \rho f, \qquad \sigma = -p_{\mathrm{phys}} I + 2\mu D(u), \quad D(u) = \tfrac{1}{2}(\nabla u + \nabla u^{\mathsf{T}}).$$

The constitutive assumptions are a Newtonian fluid, constant density $\rho > 0$, and constant dynamic viscosity $\mu > 0$, with kinematic viscosity $\nu = \mu/\rho$. Incompressibility then reduces the viscous force per unit mass to $\nu\Delta u$. In the equations used here, $p = p_{\mathrm{phys}}/\rho$ has units of velocity squared and f has units of acceleration. A non-Newtonian stress, variable density, compressibility, temperature-dependent transport, or relativistic stress requires a different model. Such extensions are not hidden in the response fields.

For a characteristic speed U and length L, use $t_* = L/U$, $p_* = U^2$, and $f_* = U^2/L$. The dimensionless physical momentum equation is

$$\partial_\tau \widehat{u} + (\widehat{u}\cdot\widehat{\nabla})\widehat{u} + \widehat{\nabla}\widehat{p} - \frac{1}{\mathrm{Re}}\widehat{\Delta}\widehat{u} = \widehat{f}, \qquad \mathrm{Re} = \frac{UL}{\nu}.$$

All momentum terms now have the same acceleration scale U^2/L. The divergence residual has scale U/L instead. Adding their raw numerical magnitudes would mix unlike units; both must

be normalized before a single stopping threshold is meaningful. The Reynolds number compares advective transport and viscous diffusion. It determines neither the action degree nor the existence of a useful error estimate by itself. The periodic unit domain below can be understood as these already scaled coordinates.

Introduce independent real fields u, p, v, q. The momentum residual and action are

$$R_m(u,p) = \partial_t u - \nu\Delta u + (u\cdot\nabla)u + \nabla p - f, \tag{2.6}$$

$$S_{\mathrm{NS}}[u,p,v,q] = \int_0^T \int_\Omega \left[v\cdot R_m(u,p) + q\,\nabla\cdot u\right] \mathrm{d}x\,\mathrm{d}t. \tag{2.7}$$

All fields are sufficiently smooth for the displayed products and integrations by parts. Pressure is normalized to spatial mean zero. Response pressure q can receive the same normalization. On a periodic domain this loses no divergence equation, since a divergence already has zero spatial integral.

For this response-field action use the dimensionless coordinates and physical fields scaled above, and take v, q dimensionless as well. The displayed integral is then dimensionless. Multiplication by a fixed action scale S_0 restores joule seconds without changing its stationary equations. The response fields are multiplier amplitudes enforcing the balances; they have not been assigned an elastic energy or a material constitutive law. This choice makes both the scaling and the saddle character of the construction explicit.

Variations in v and q give, respectively, $R_m = 0$ and $\nabla\cdot u = 0$. Variation in p gives $\nabla\cdot v = 0$. To obtain the remaining equation, vary u before imposing incompressibility. The nonlinear derivative is

$$\delta\big((u\cdot\nabla)u\big) = (\delta u\cdot\nabla)u + (u\cdot\nabla)\delta u.$$

The second term transfers a derivative onto uv; its divergence includes an off-shell contribution $-(\nabla\cdot u)v$. The complete interior stationary system is consequently

$$\begin{aligned} &R_m(u,p) = 0, \qquad \nabla\cdot u = 0, \qquad \nabla\cdot v = 0, \\ &-\partial_t v - \nu\Delta v - (u\cdot\nabla)v - (\nabla\cdot u)v + (\nabla u)^{\mathsf{T}} v - \nabla q = 0. \end{aligned} \tag{2.8}$$

Here $(\nabla u)_{ij} = \partial_j u_i$. On the physical constraint surface the divergence contribution vanishes, leaving the usual formal adjoint of the momentum linearization, coupled to response pressure.

For initial-value data impose $u(0) = u_0$, so $\delta u(0) = 0$. The time integration produces the terminal term $\int_\Omega v(T)\cdot\delta u(T)\,\mathrm{d}x$. Allowing the final velocity to vary freely gives the natural condition $v(T) = 0$. Periodicity removes all spatial boundary terms. These choices completely specify the boundary sector used in the following result.

Proposition 2.2 (Exact stationary projection and lift)**.** *Every sufficiently regular stationary point of* (2.7)*, with these data, projects to a solution* (u,p) *of incompressible Navier–Stokes. Every sufficiently regular such solution has a stationary lift* $(u,p,0,0)$*. The enlarged action has* $D^4 S_{\mathrm{NS}} = 0$*.*

Proof. The response variations impose the physical equations. Conversely zero response satisfies its divergence, adjoint, and terminal equations, while the physical solution satisfies the other two variations. The advective term $v_i u_j \partial_j u_i$ is cubic. Time, viscosity, pressure, and divergence terms are quadratic, and the fixed-force term is linear. Theorem 1.1 therefore applies. □

The response velocity v and response pressure q are independent multipliers for momentum and incompressibility. They are not a second physical fluid, a measured negative viscosity, or hidden turbulent fluctuations. Their scales can be chosen so that the two pairings in the normalized action are dimensionless. Rescaling a multiplier by a fixed nonzero factor does not alter the equation it enforces, but it does change the numerical weighting of its assembled residual. That weighting must therefore be reported when a code compares momentum and divergence errors. In adjoint-based sensitivity calculations, nonzero response fields are instead driven by a selected objective and its terminal data; the zero-terminal-data stationary lift proved here has no such objective term.

The response fields impose equations by stationarity; they do not turn viscous flow into a positive action minimum. At zero response the action is zero even for an off-shell velocity. When a constraint residual pairs nontrivially with a response variation, scaling that variation changes the action with either sign. Positivity and coercivity cannot be inferred from its cubic degree.

Eliminating responses illustrates why the independent-field qualification matters. In a dimensionless finite realization with a fixed inner product, replace the action by

$$S_\epsilon = \langle v, R_m \rangle + \langle q, \nabla \cdot u \rangle - \frac{\epsilon}{2}(\|v\|^2 + \|q\|^2), \qquad \epsilon > 0.$$

Its response equations give $v = R_m/\epsilon$ and $q = (\nabla \cdot u)/\epsilon$. Substitution yields

$$S_\epsilon^{\mathrm{red}}(u, p) = \frac{1}{2\epsilon}\left(\|R_m(u, p)\|^2 + \|\nabla \cdot u\|^2\right).$$

The enlarged penalized action remains cubic, but this reduced objective is generally quartic. Its stationary equations impose an adjoint residual condition, not automatically a zero residual. Consequently a penalized stationary point is not interchangeable with the exact stationary projection proved above. This finite calculation identifies the precise operation responsible for the change of derivative order; it makes no limiting assertion as ϵ tends to zero.

Physical dissipation instead follows directly from the momentum equation. For a smooth periodic solution, multiplying by u and integrating gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{L^2}^2 + \nu\|\nabla u\|_{L^2}^2 = \int_\Omega f \cdot u \,\mathrm{d}x. \tag{2.9}$$

Incompressibility cancels advection and pressure work. This is a kinetic energy balance, distinct from minimization of the response action. It does not by itself control every higher derivative. Its use in an error estimate requires the regularity and stability assumptions of that estimate, rather than an inference from the polynomial closure alone.

For numerical work the energy balance is a specific diagnostic. In dimensional form the fluid kinetic energy is $\rho \int |u|^2 \,\mathrm{d}x/2$, viscous conversion is $\mu \int |\nabla u|^2 \,\mathrm{d}x$ on this periodic incompressible domain, and forcing contributes $\rho \int f \cdot u \,\mathrm{d}x$. A discrete convective operator that does not preserve the required skew-adjoint cancellation can create an energy defect even when its assembled nonlinear residual is small. Pressure requires a compatible velocity–pressure discretization or an exact divergence-free representation; an arbitrary pressure mode can be invisible to the momentum test. These issues are controlled by the discretization and its stability constants, not by adding more response variables.

The two familiar time scales also remain: diffusion acts over L^2/ν, while transport acts over L/U. On a resolved Fourier mode, viscosity contributes decay rate $\nu|k|^2$, and advection about a velocity of order U contributes frequencies of order $U|k|$. An explicit integrator must respect the stability restrictions of the actual spatial operator. An implicit Newton solve trades those restrictions for a

conditioned nonlinear and linear algebra problem; it does not delete the physical decay or phase error. The periodic Taylor–Green calculation in Section 5.4 gives exact velocity, pressure, decay, and energy values against which these parts can be checked.

2.4 Bare QED on a fixed background

Choose Minkowski spacetime with $\eta = \mathrm{diag}(1,-1,-1,-1)$, natural units $\hbar = c = 1$, and fixed gamma matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I$ and $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$. The gauge potential A_μ is real and commuting; ψ and $\bar{\psi}$ are independent anticommuting spinor variables in the independent-field variation. The physical conjugacy is $\bar{\psi} = \psi^\dagger\gamma^0$. We use a trivial gauge bundle and compactly supported variations, so a single potential and the following integrations by parts are available. Tong's Sections 6.1 and 6.3.1 fix the Maxwell and charged-fermion construction [20].

In these rationalized natural units the action is dimensionless, energy and inverse length have the same units, and the local Lagrangian density has units of energy to the fourth power. Dimensional counting gives $[A_\mu] = \text{energy}$, $[\psi] = [\bar{\psi}] = \text{energy}^{3/2}$, $[m] = \text{energy}$, and dimensionless coupling e. A reference length L therefore permits $\widehat{A} = LA$, $\widehat{\psi} = L^{3/2}\psi$ and $\widehat{x} = x/L$. The parameters of the scaled bare equations are e and mL. Restoring $\hbar, c$ makes the latter L/λ_C, where $\lambda_C = \hbar/(mc)$ is the reduced Compton wavelength. In the usual SI convention the fine-structure coupling is $\alpha = e_{\mathrm{SI}}^2/(4\pi\epsilon_0\hbar c)$. This conversion is necessary before comparing a dimensionless spinor calculation with an energy or cross-section measurement.

Define

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad D_\mu\psi = (\partial_\mu + ieA_\mu)\psi,$$

and take the bare classical action

$$S_{\mathrm{QED}} = \int \left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu A_\mu\psi\right] \mathrm{d}^4x. \tag{2.10}$$

Using left and right fermionic variations consistently gives the two Dirac equations

$$(i\gamma^\mu D_\mu - m)\psi = 0, \qquad i(D_\mu^{\leftarrow}\bar{\psi})\gamma^\mu + m\bar{\psi} = 0, \tag{2.11}$$

where $D_\mu^{\leftarrow}\bar{\psi} = \partial_\mu\bar{\psi} - ieA_\mu\bar{\psi}$. For the gauge variation, antisymmetry gives $\delta(-F^2/4) = -F^{\mu\nu}\partial_\mu\delta A_\nu$. After integration by parts its coefficient is therefore

$$\partial_\mu F^{\mu\nu} = e\bar{\psi}\gamma^\nu\psi. \tag{2.12}$$

The homogeneous Maxwell equation $\partial_{[\lambda}F_{\mu\nu]} = 0$ follows from $F = \mathrm{d}A$, without another multiplier. Equations (2.11) imply conservation of $j^\mu = \bar{\psi}\gamma^\mu\psi$: substituting the two Dirac equations in $\partial_\mu j^\mu$ cancels both the mass and potential terms. Thus the source in (2.12) satisfies the compatibility condition obtained by taking its divergence.

The $\nu = 0$ Maxwell equation is Gauss's constraint $\partial_i F^{i0} = e\psi^\dagger\psi$. It remains part of the stationary system when the spatial components are used as evolution equations. In a Lorenz gauge, $\partial_\mu A^\mu = 0$, the potential equation takes the form $\Box A^\nu = ej^\nu$; that gauge condition is an additional choice of representative, not another equation obtained by varying an independent multiplier in the displayed action.

A numerical state for the electromagnetic sector may contain the potential, the electric and magnetic fields, or both. The physical fields are derivatives of the potential; checking only a potential residual

can miss a derivative error unless the chosen norm controls it. Gauss's law is a constraint on the initial fields and charge, and charge continuity makes its propagation compatible with Maxwell evolution. A discrete current deposition that violates continuity can therefore produce a Gauss-law error independently of the formal action degree. A boundary with an incoming electromagnetic wave, a conducting wall, and a periodic boundary impose different energy fluxes. Their treatment must be fixed before declaring energy drift to be a solver defect.

The convention also fixes the gauge action: $A_\mu \mapsto A_\mu + \partial_\mu \alpha$, $\psi \mapsto e^{-ie\alpha}\psi$, and $\bar{\psi} \mapsto \bar{\psi} e^{ie\alpha}$. These transformations preserve the action and current. Gauge-equivalent potentials describe the same electromagnetic field; their redundant coefficients leave zero modes in the unconstrained tangent matrix. A chosen gauge and constraint treatment are consequently part of any numerical stability claim.

Maxwell and free Dirac terms are quadratic; the ordered product $\bar{\psi} A \psi$ is cubic. With a commuting affine parameter s, substitution of $(A, \psi, \bar{\psi}) + s(a, \chi, \bar{\chi})$ produces a polynomial of degree at most three, coefficient by coefficient in the Grassmann algebra. Hence $D^4 S_{\mathrm{QED}} = 0$ formally, and the Euler–Lagrange residual in this independent-field presentation is at most quadratic. This calculation does not assign an ordinary positive norm to Grassmann numbers.

The words *bare QED* delimit what this test controls. The Maxwell–Dirac action is the classical polynomial used to define the interactions; QED predictions also involve a regulator, quantum states, and a prescription for determining renormalized parameters from physical measurements. The bare mass symbol m need not equal a measured pole mass before that prescription is supplied. A tree-level interaction term contains three fields, but a scattering probability contains products and sums of amplitudes, and loop corrections introduce integrations over internal momenta. Their cutoff dependence and physical uncertainties are not determined by differentiating the bare action four times.

For a regulated coefficient calculation one may choose a finite ordered basis of fermion monomials and compare the resulting coefficient arrays. This supplies ordinary numerical norms for a particular finite representation while retaining the anticommutation rule. It is not a probability norm on a Grassmann-valued spinor. For a quantum calculation, state normalization, positive probabilities, and the observable's error estimate belong to the quantum construction itself. The separate vacuum Maxwell mode and free Dirac operator checks in Section 5.5 test parts of a QED implementation without pretending that either one alone verifies an interacting quantum prediction.

One can check the coefficient count without commuting any fermion factors. Keep the order $\bar{\psi} A \psi$ fixed while expanding the three affine factors. The highest power is $-es^3 \bar{\chi} \gamma^\mu a_\mu \chi$; every other contribution has lower degree. Anticommutation signs matter when differentiating with respect to fermionic coordinates, but they cannot create a fourth factor or a fourth power of the commuting parameter. This is the precise formal content of the closure statement for fermions. It is stronger than a cancellation in one numerical example, while remaining distinct from a convergence theorem for a quantum expectation value.

Quantization changes the objects being compared: quantum states, correlation functions, and renormalized observables require a specified construction and regulator. The bare degree identity persists for the displayed polynomial; an effective action obtained by integrating out fields need not be cubic. A fixed curved spin background retains this count when its connection and volume element are coefficients. Making that background dynamical, using nonlinear link variables, or changing the boundary sector requires a fresh count in the newly declared fields. The Abelian qualification also matters: here F is linear in A. In a non-Abelian potential formulation the additional $[A_\mu, A_\nu]$

term makes F^2 contain quartic monomials. The argument for bare QED does not transfer to that presentation by changing the name of the gauge field.

2.5 The common conclusion and its domain

Theorem 2.3 (Shared closure of the selected presentations)**.** *In the regular boundary sectors specified above, the independent-field actions satisfy $D^4 S_i = 0$ and their Euler–Lagrange residuals satisfy $D^3 R_i = 0$. For QED these are coefficient identities in the independent-field fermion presentation. For Schrödinger the stronger identities $D^3 S = 0$ and $D^2 R = 0$ hold.*

Proof. Each action has the displayed polynomial degree, and variation lowers that degree by one. Apply Theorem 1.1 in the respective field coordinates; the same coefficient argument works with anticommuting fermion coefficients and a commuting affine parameter s. □

The result survives a fixed affine discretization. If $Z = Z_0 + E_h z$, with E_h linear and independent of z, then $S_{i,h}(z) = S_i(Z_0 + E_h z)$ still has degree at most three. Fixed linear quadrature preserves the count. With the inherited dual pairing its residual is $E_h^* R_i(Z_0 + E_h z)$, of degree at most two. This gives an exact discrete algebraic conclusion without assuming that the discrete stationary fields approximate the continuum fields. State-dependent quadrature, a nonlinear coordinate map, or elimination of discrete auxiliaries falls outside this particular preservation argument. Approximation and stability require their own estimates even when the discrete degree identity is exact.

The field spaces remain different. Lorentzian metrics occupy a nondegenerate sector, wavefunctions live in a complex Hilbert space, incompressible velocities satisfy a differential constraint, and QED includes gauge and spinor structure. Their initial data, boundary conditions, and physical solutions are not identified by the common degree. The theorem supplies an exact shared derivative hierarchy. Comparing their readings requires the declared observation maps and stability bounds developed in the following chapter.

Chapter 3

From closure to an engineering tolerance

An engineering tolerance is an allowable change in a stated quantity: a length, a force, a phase, a mass-flow rate, or a reported number. Its value comes with units, an operating range, a reference state, and a method of comparison. The fourth-order closure proved in the preceding chapters removes one Taylor remainder. It supplies neither the units of a measurement nor the sensitivity of an equilibrium to disturbances. This chapter supplies the intermediate calculations needed to turn a residual into a finite allowance on an observable.

3.1 Put the equations and their tolerances in compatible units

Let Z collect the unknown coefficients of a selected spatial and temporal discretization. Some coefficients can represent displacements, others velocities, pressures, potentials, or multiplier reactions. The assembled equation is $R(Z; d) = 0$, where d contains the prescribed loads, material coefficients, boundary values, and initial data. Its components can have different dimensions. Adding a force residual in newtons to a continuity residual in cubic metres per second has no physical meaning.

Choose positive diagonal scale matrices D_Z and D_R, and a reference state Z_0. Define dimensionless state and residual coordinates by

$$z = D_Z^{-1}(Z - Z_0), \qquad r(z; d) = D_R^{-1} R(Z_0 + D_Z z; d). \tag{3.1}$$

Every entry of D_Z has the units of its state component; every entry of D_R has the units of its residual component. The dimensionless Jacobian is

$$J(z) = D_R^{-1} \frac{\partial R}{\partial Z}(Z_0 + D_Z z) D_Z.$$

A threshold on $\|r\|_2$ is consequently a statement about specified component scales. It is reproducible only if those scales are recorded. For example, dividing force imbalance by a one-newton allowable imbalance and volume imbalance by a $10^{-8}\,\mathrm{m^3\,s^{-1}}$ allowance gives both equations a dimensionless interpretation. An order-one residual then has a clear meaning relative to the selected allowances.

A raw Euclidean norm also changes when cells or finite-element nodes are added. For values of a

scalar field sampled on cells with volumes V_i, a useful normalized root-mean-square norm is

$$\|e\|_{\mathrm{rms}}^2 = \frac{\sum_i V_i (e_i/e_{\mathrm{scale}})^2}{\sum_i V_i}.$$

This approximates a volume-averaged continuum quantity under mesh refinement. A mass matrix supplies the corresponding finite-element quadrature. A maximum norm serves a different requirement: it protects the largest component rather than an average. Thus an RMS temperature tolerance cannot by itself certify the hottest point. Below, the norms and singular values refer to the declared scaled coordinates; a weighted Euclidean norm can be converted to ordinary Euclidean coordinates by its positive square root. This is numerical scaling with an explicit physical interpretation, not a new physical law [10, 14].

For incompressible flow, characteristic speed U, length L, density ρ, and viscosity μ give time scale L/U and inertial pressure scale ρU^2. Dividing momentum by $\rho U^2/L$ produces a viscous coefficient $1/\mathrm{Re}$, where $\mathrm{Re} = \rho UL/\mu$. These are physical nondimensionalization choices. They are distinct from selecting a residual threshold to meet, for example, a ten-micrometre displacement allowance. At very small Reynolds number a viscous pressure scale can be more useful numerically. Changing a scale changes a matrix condition number and the interpretation of its norm; it does not change the underlying dimensional balance or authorize a different tolerance.

3.2 Exact remainder and Jacobian-free evaluation

For the quadratic residual in (1.2), Taylor expansion terminates exactly:

$$R(Z+H) - R(Z) - DR(Z)H = B(H,H) = \tfrac{1}{2}D^2R[H,H]. \tag{3.2}$$

If $\|D^2R\| \le M_2$ in specified norms, the linearization defect is at most $M_2\|H\|^2/2$. The missing fourth action term gives a quadratic residual defect. A Newton correction still has a finite defect because a quadratic equation is not a linear equation.

For $\delta \neq 0$,

$$\frac{R(Z+\delta H) - R(Z)}{\delta} = DR(Z)H + \frac{\delta}{2}D^2R[H,H]. \tag{3.3}$$

Both bilinear slots in (1.3) are present. A centered difference gives exactly $DR(Z)H$ in exact arithmetic because the quadratic terms cancel. If each residual evaluation has error at most e_R, the forward-difference error is bounded by

$$\tfrac{1}{2}|\delta| M_2 \|H\|^2 + \frac{2e_R}{|\delta|}.$$

The two terms have opposite responses to decreasing δ: truncation falls while cancellation and evaluation error are amplified. A matrix-free calculation must budget both. Chapter 4 uses this bound to choose a differencing scale and an achievable inner linear tolerance [15].

Suppose a linear solver returns $DR(Z)H = -R(Z) + \ell$. Equation (3.2) gives

$$\|R(Z+H)\| \le \|\ell\| + \tfrac{1}{2}M_2\|H\|^2.$$

The first term is an incomplete linear solve; the second is nonlinear curvature over the proposed step. Neither is a mesh error. Preconditioning changes a third quantity: if P is invertible, a reported $\|P^{-1}R\| \le \eta$ implies only $\|R\| \le \|P\|\,\eta$ in compatible norms. Always convert the monitored norm to the residual norm used by the physical acceptance test.

3.3 A sharp cubic interpolation certificate

Let a real $\phi \in C^4([a,b])$ be interpolated by the cubic $H_3\phi$ matching value and first derivative at both endpoints; write $h = b - a$. The Hermite remainder is

$$\phi(x) - H_3\phi(x) = \frac{\phi^{(4)}(\xi_x)}{4!}(x-a)^2(x-b)^2.$$

To obtain it, subtract a suitable multiple of $(t-a)^2(t-b)^2$ from the interpolation error, make the result vanish at x, and apply the repeated Rolle argument, counting both endpoint zeros twice. Since the maximum of the product is $h^4/16$,

$$\|\phi - H_3\phi\|_\infty \le \frac{h^4}{384}\left\|\phi^{(4)}\right\|_\infty. \tag{3.4}$$

For $\phi(x) = x^4$ on $[0,1]$, the error is $x^2(1-x)^2$ and reaches $1/16 = 24/384$. The constant is sharp for this class.

This estimate has a direct dimensional reading. If ϕ is a transverse beam displacement measured in metres and x is measured in metres, then $\phi^{(4)}$ has units m^{-3}. Multiplication by h^4 restores a displacement. For a prismatic Euler–Bernoulli beam with constant bending stiffness EI and distributed transverse load f, the static equation is $EI\phi^{(4)} = f$. On an interval where $|f| \le f_{\max}$, a certified bound is

$$\|\phi - H_3\phi\|_\infty \le \frac{h^4 f_{\max}}{384EI}.$$

Here EI is measured in force times length squared. A target interpolation allowance ϵ_b therefore admits $h \le (384EI\epsilon_b/f_{\max})^{1/4}$ when $f_{\max} > 0$. An unloaded constant-stiffness span is cubic and has zero interpolation error under these endpoint data. Variable stiffness instead gives $(EI\phi'')'' = f$ and requires its own derivative bound. Point loads and stiffness jumps require intervalwise estimates and junction conditions.

The same remainder applies to a smooth calibration trace plotted against a dimensionless sweep parameter. It concerns a physical displacement curve or a scalar readout only when its own fourth derivative is bounded. A nonlinear readout can turn a cubic state trace into a noncubic output. The off-shell action's zero fourth variation cannot be substituted for a measured displacement derivative without deriving that substitution [4, 5].

3.4 Sensitivity through the smallest Jacobian singular value

A small force imbalance implies a small displacement only for a system with sufficient stiffness. The numerical version of that statement is a lower singular-value bound on the scaled tangent matrix. A nearly free mechanism can move a large distance while producing a tiny restoring force.

Proposition 3.1 (Conditional residual-to-state bound)**.** *Let z_* solve the finite scaled equations $r(z_*) = 0$. Suppose the smallest singular value of $J_* = Dr(z_*)$ is at least $\sigma > 0$, and along the straight segment from z_* to z,*

$$\|Dr(z_* + te) - J_*\|_2 \le Lt\,\|e\|_2, \qquad e = z - z_*, \quad 0 \le t \le 1.$$

If $L\,\|e\|_2 \le \sigma$, then

$$\|z - z_*\|_2 \le \frac{2}{\sigma}\,\|r(z)\|_2.$$

Equivalently the stability constant is $K = 1/\sigma$ and the localization condition is $KL\|e\|_2 \le 1$.

Proof. The fundamental theorem of calculus along the segment gives $r(z) = J_* e + \rho$ with $\|\rho\|_2 \le L\|e\|_2^2/2$. The defining singular-value inequality gives $\sigma\|e\|_2 \le \|J_* e\|_2 \le \|r(z)\|_2 + L\|e\|_2^2/2$. Under the stated localization condition the quadratic term is at most $\sigma\|e\|_2/2$, which can be moved to the left. □

This is a local estimate about an established equilibrium and an identified region. A validated Newton calculation, a direct solution, or a separate existence argument must supply those premises. The residual alone does not prove them. In the one-component spring equation $r_\epsilon(z) = \epsilon z$, the trial displacement $z = 1$ has residual ϵ and error one. The smallest tangent singular value is exactly ϵ, so the estimate correctly deteriorates as the spring becomes soft.

For a measured output with local gradient $g = \nabla_z Q$, linear sensitivity is $g^\mathsf{T} J_*^{-1} r$. This can be far smaller than the worst-case state estimate $\|g\|_2 \|J_*^{-1}\|_2 \|r\|_2$. Solving the adjoint system $J_*^\mathsf{T}\lambda = g$ evaluates the first-order output defect as $\lambda^\mathsf{T} r$. Any use as a rigorous error bound still needs the nonlinear remainder and localization allowance. An adjoint estimate is particularly useful when a pressure drop is required accurately but every velocity coefficient need not have the same absolute accuracy.

3.5 Pressure and multiplier constraints are part of the solve

A linearized constrained equilibrium often has the block form

$$\begin{bmatrix} A & B^\mathsf{T} \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix}. \tag{3.5}$$

Here u is the primary field, p is a multiplier, and $Bu = g$ is a constraint such as discrete volume conservation. The sign convention for p has been selected so the upper off-diagonal block is B^T. When A is invertible, elimination gives

$$Sp = BA^{-1}f - g, \qquad S = BA^{-1}B^\mathsf{T}, \qquad u = A^{-1}(f - B^\mathsf{T}p).$$

The Schur complement S expresses how a pressure correction changes the constraint after the velocity response. If A is symmetric positive definite and B has full row rank, then $p^\mathsf{T}Sp = (B^\mathsf{T}p)^\mathsf{T}A^{-1}(B^\mathsf{T}p) > 0$ for nonzero p. A small eigenvalue of S signals a pressure pattern that hardly changes the constraint and therefore can be poorly determined.

An elementary example makes the reaction explicit. Take $A = \mathrm{diag}(2,2)$, $B = [1 \;\; 1]$, $f = (2,0)^\mathsf{T}$, and $g = 0$. Then $S = 1$, $p = 1$, and $u = (1/2, -1/2)^\mathsf{T}$. Ignoring pressure would give $u = (1,0)^\mathsf{T}$ and violate $Bu = 0$. Accurate momentum equations alone would consequently accept the wrong constrained state.

For incompressible flow with closed or periodic boundary conditions, pressure can be determined only up to a constant. Fix its mean or one reference value before asking for an invertible matrix. Compatibility of the prescribed boundary flux is also required: an enclosed volume cannot have a nonzero net prescribed inflow in an incompressible steady problem. Discrete velocity and pressure spaces must retain a positive inf-sup constant in their selected mass or energy norms. That condition prevents an undetected pressure pattern. In finite coordinates it amounts to a lower singular-value bound for the appropriately scaled constraint operator on the pressure space after its constant mode has been removed. A multiplier variable alone supplies neither that bound nor flux compatibility [3].

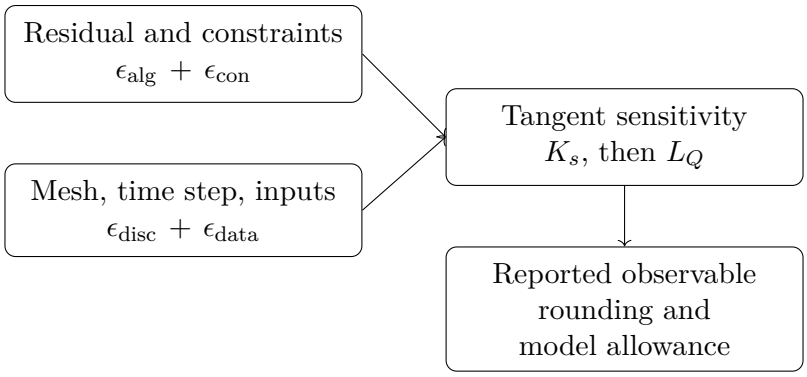


Figure 3.1: A physical tolerance passes through the actual operators and scales. Polynomial closure controls a remainder; the remaining allowances retain separate measurements or estimates.

3.6 An error budget for a physical reading

Suppose a selected model, its mesh transfers, and its stability estimates supply

$$\|Z_h - Z_*\|_X \le K_s(\epsilon_{\text{alg}} + \epsilon_{\text{con}} + \epsilon_{\text{disc}} + \epsilon_{\text{data}}). \tag{3.6}$$

Here the state is compared in one declared normalized norm. ϵ_{alg} includes the incomplete nonlinear and linear solves and residual evaluation error; ϵ_{con} measures constraint defects after compatible scaling. ϵ_{disc} measures spatial and temporal consistency defects in the same residual units. ϵ_{data} measures changes in equations induced by calibrated input allowances, including loads and constitutive parameters. Writing the sum assumes a deterministic enclosure for each contribution. A statistical standard uncertainty is a different object and is not an absolute error bound without additional coverage assumptions [12, 14].

If Q changes by at most L_Q times the state change in this region and its evaluation and reporting error is at most ϵ_Q, then

$$\left|\widehat{Q} - Q(Z_*)\right| \le L_Q K_s \sum_j \epsilon_j + \epsilon_Q. \tag{3.7}$$

Insert $Q(Z_h)$ between the two outputs and apply the triangle inequality. For $K_s = 2$, $L_Q = 3$, residual-unit allowances totaling 10^{-6}, and readout allowance 10^{-7}, the output bound is $6.1 \cdot 10^{-6}$ in the chosen output units. The common derivative order determined none of those sensitivity constants.

The selected equation can itself disagree with a physical experiment. If a separately supported model-discrepancy allowance satisfies $|Q(Z_*) - Q_{\text{physical}}| \le \epsilon_{\text{model}}$, then add ϵ_{model} to the right side. Refining a mesh does not calibrate this term. Nor does driving an algebraic residual to machine precision establish an uncertain material coefficient. Calibration connects an indication to a reference quantity through a documented measurement model [13].

3.7 Two readers, decimal decisions, and cycle closure

If two instruments or two numerical representations estimate the same quantity q with certified allowances E_1, E_2, then

$$|\widehat{q}_1 - \widehat{q}_2| \le E_1 + E_2.$$

If they estimate different quantities q_1, q_2, add a bound on $|q_1 - q_2|$. Agreement within the combined allowance is a consistency test of the declared common measurement. It cannot establish the measurement model without that declaration and its calibration evidence. Conversely disagreement larger than the sum rejects at least one allowance, the common-quantity assumption, or the implementation.

A reporting cell adds a discontinuous final decision. For truncation with resolution $\Delta > 0$, cell k is $[k\Delta, (k+1)\Delta)$. A certified closed interval wholly within one cell fixes the reported integer k. A narrow interval crossing its upper edge does not. A nearest-rounding rule has shifted boundaries and needs a declared tie rule. Chapter 6 checks the relevant endpoints by exact integer arithmetic instead of treating a repeated display as a certificate.

Convergence also needs its target. A Cauchy sequence of residuals need not converge to zero: $1+1/n$ is an elementary counterexample. If residual and consistency allowances ρ_n tend to zero, a common stable reference state exists, and one uniform constant gives $\|Z_n - Z_*\| \le K\rho_n$, then the computed states converge to that state. On changing meshes the comparison requires explicit transfers to one physical norm. A decreasing relative residual is insufficient if its normalizing load or the stability constant is changing without control.

Finally, cycle closure is a measured endpoint condition. For an exact reading $q(t)$ and computed reading $\widehat{q}(t)$ whose two endpoint errors are at most ϵ, the triangle inequality gives

$$\|\widehat{q}(1) - \widehat{q}(0)\| \le \|q(1) - q(0)\| + 2\epsilon.$$

An exact closed cycle therefore admits the finite closure allowance 2ϵ. A cubic trace need not close: $\phi(s) = s^3$ has $\phi^{(4)} = 0$ and endpoint difference one. Cycle closure, residual convergence, and the resolution of successive recorded events consequently require separate engineering acceptance tests.

Chapter 4

Newton–Krylov solution and multigrid correction

A nonlinear equation, its linearized correction, and the algorithm used to compute that correction are three different engineering objects. This chapter follows an incompressible-flow time step through all three. The same numerical organization can be used for other residual equations, provided their fields, constraints, boundary conditions, and tangent operators are supplied explicitly. A short recurrence at the bottom of a solver does not determine the differential equation that produced its matrix [15, 1].

4.1 Start with the momentum and volume balances

For density $\rho > 0$, viscosity $\mu > 0$, and force per unit volume f, backward Euler applied to incompressible momentum gives the residual

$$R_m(u, p) = \rho \frac{u - u^{\text{old}}}{\Delta t} + \rho(u \cdot \nabla)u + \nabla p - \mu \Delta u - f, \tag{4.1}$$

$$R_c(u) = \nabla \cdot u. \tag{4.2}$$

The spatial derivatives here denote the declared discrete operators when the equations are assembled. Momentum components have units of force per volume; divergence has units of inverse time. They require separate residual scales as in (3.1). Boundary fluxes, velocity values, the previous time state, and a pressure reference are part of the equation specification. They cannot be recovered from a Krylov iteration history.

For a velocity correction v and pressure correction π, the tangent is

$$\begin{aligned} DR_m(u, p)[v, \pi] &= \rho v/\Delta t + \rho(v \cdot \nabla)u + \rho(u \cdot \nabla)v + \nabla \pi - \mu \Delta v, \\ DR_c(u)[v] &= \nabla \cdot v. \end{aligned} \tag{4.3}$$

Both convection terms are required. Expanding $(u + v) \cdot \nabla(u + v)$ leaves the exact nonlinear defect $\rho(v \cdot \nabla)v$. Freezing the advecting velocity retains only $(u \cdot \nabla)v$; that produces a useful Picard or Oseen iteration, but a different tangent from Newton's method. For a fixed discrete bilinear convection operator the same expansion is an exact coefficient identity. Limiters, changing turbulence

closures, contact switches, or state-dependent mesh transformations can change the polynomial and differentiability assumptions and must be included in the residual actually differentiated.

Write all scaled equations as $r(z) = 0$. At nonlinear iteration k, Newton asks for a correction s_k satisfying

$$J_k s_k = -r_k, \qquad J_k = Dr(z_k), \quad r_k = r(z_k).$$

The next sections explain how to approximate this solve, quantify the approximation, and decide whether the resulting physical reading meets its tolerance.

4.2 A matrix-free tangent has a finite error floor

A Jacobian-free product uses residual evaluations in place of storing J_k. For a unit direction v in the scaled state norm, define

$$\widehat{Jv} = \frac{\widehat{r}(z + \delta v) - \widehat{r}(z)}{\delta}.$$

Assume each evaluated residual differs from its exact discrete value by at most e_R, and $\|D^2 r[v, v]\| \leq M_2$ over the step. Then

$$\left\| \widehat{Jv} - Jv \right\| \leq \frac{M_2 \, |\delta|}{2} + \frac{2e_R}{|\delta|}. \tag{4.4}$$

For a quadratic residual this truncation expression follows from an exact Taylor identity. For a general smooth residual it follows by integrating the second derivative. The residual-evaluation allowance includes floating-point roundoff and any inexact subsidiary calculation. It must be supplied or estimated for the implementation; setting it equal to machine precision without accounting for residual magnitudes is not a bound.

When $M_2, e_R > 0$, minimizing the right side gives

$$|\delta_{\text{opt}}| = 2\sqrt{e_R/M_2}, \qquad \epsilon_{Jv,\min} = 2\sqrt{M_2 e_R}.$$

Thus an arbitrarily small differencing displacement does not give an arbitrarily accurate tangent product. For a nonunit direction, apply the formula to $v/\|v\|$ and multiply the result by $\|v\|$, keeping the state displacement at the selected size. Component scales in D_Z make that displacement dimensionally meaningful. A direction dominated by pressure otherwise could select a completely different physical perturbation from one dominated by velocity.

For a quadratic residual the centered quotient has no truncation defect in exact arithmetic, but its evaluation-error bound is $e_R/|\delta|$. For a general residual with third-derivative bound M_3, the corresponding bound is $M_3\delta^2/6 + e_R/|\delta|$. Neither estimate licenses a step outside the admissible material range or across a nonsmooth switch. An analytic tangent, automatic differentiation of the actual residual, or a carefully validated difference quotient can reduce this uncertainty; all still require finite arithmetic analysis [15].

4.3 Inexact Newton: spend accuracy where it changes the answer

Solving every early linear system to the final reporting tolerance wastes work while the nonlinear state is still far away. An inexact Newton step requires instead

$$\|J_k s_k + r_k\| \leq \eta_k \|r_k\|, \qquad 0 \leq \eta_k < 1. \tag{4.5}$$

The dimensionless forcing number η_k is the fraction of the current residual left by the linearized solve. Combining this inequality with the second-derivative remainder gives

$$\|r(z_k + s_k)\| \le \eta_k \|r_k\| + \tfrac{1}{2} M_2 \|s_k\|^2 . \tag{4.6}$$

If $\left\|J_k^{-1}\right\| \le K$, then $\|s_k\| \le K(1+\eta_k)\|r_k\|$, so the right side is at most

$$\eta_k \|r_k\| + \tfrac{1}{2} M_2 K^2 (1+\eta_k)^2 \|r_k\|^2 .$$

A fixed forcing fraction retains a linear term. Choosing $\eta_k \le c\|r_k\|$ in a sufficiently small stable neighborhood makes both terms quadratic in the residual. Such a choice is useful only above the arithmetic and tangent-product error floors.

A Krylov monitor typically controls the computed linearized residual. If the product approximation acts as $J_k + E_k$, and the computed solve has residual at most $\widehat{\eta}_k \|r_k\|$, then the true tangent residual is bounded by

$$\|J_k s_k + r_k\| \le \widehat{\eta}_k \|r_k\| + \|E_k s_k\| .$$

For direction-dependent or varying product errors, the accumulated Arnoldi relation has to be bounded instead of assuming one fixed matrix E_k. Either way, asking the computed inner residual to fall far below an uncontrolled product error does not establish (4.5).

A full Newton step can leave the local stable region. A line search uses $z_{k+1} = z_k + \lambda s_k$, with $0 < \lambda \le 1$. For $F(z) = \|r(z)\|_2^2/2$, (4.5) gives

$$DF(z_k)[s_k] = r_k^{\mathsf{T}} J_k s_k \le -(1-\eta_k)\|r_k\|_2^2 .$$

Thus the correction is a strict descent direction when $r_k \neq 0$. For any $0 < c < 1$, differentiability implies that sufficiently small positive λ satisfies the Armijo condition

$$F(z_k + \lambda s_k) \le F(z_k) - c\lambda(1-\eta_k)\|r_k\|_2^2 .$$

A practical backtracking search tries $1, 1/2, 1/4, \ldots$ while checking state admissibility and this decrease. This establishes a local acceptance mechanism, not global existence of a physical solution. If finite residual evaluation has a known enclosure, the two sides must be compared with their corresponding allowances.

4.4 Krylov vectors approximate a correction, not a field theory

For a linear correction equation $As = b$ and initial correction zero, a Krylov method searches combinations of $b, Ab, A^2b, \ldots$. GMRES constructs an orthonormal basis V_m with the Arnoldi relation

$$AV_m = V_{m+1}\overline{H}_m,$$

and selects $s_m = V_m y_m$ by minimizing $\left\|\|b\|_2 e_1 - \overline{H}_m y\right\|_2$. The small least-squares residual is the full residual norm in exact arithmetic because V_{m+1} has orthonormal columns. Loss of orthogonality, inexact products, or restarting can change the achieved reduction and must be monitored [10, 21, 1].

A preconditioner approximates the inverse response of the physical operator. With a fixed invertible right preconditioner M, solve $AM^{-1}y = b$ and recover $s = M^{-1}y$. The monitored residual remains $b - As$. Left preconditioning instead minimizes a transformed residual, whose norm must be

converted before applying a physical acceptance test. If inner preconditioner applications vary, a flexible Krylov construction is required to retain the appropriate basis-and-update relation.

Lanczos is available when the matrix is symmetric in the inner product being used. Symmetry collapses Arnoldi to

$$Av_j = \beta_{j-1}v_{j-1} + \alpha_j v_j + \beta_j v_{j+1}.$$

The older coefficients vanish because symmetry transfers each inner product $v_i^{\mathsf{T}} A v_j$ to $(Av_i)^{\mathsf{T}} v_j$, and the latter lies in an already constructed span. Conjugate gradients adds positive definiteness; a symmetric indefinite saddle matrix does not meet that condition. MINRES or an appropriate block formulation can use symmetry without positive definiteness. The convection terms in (4.3) generally make the tangent nonsymmetric, so GMRES is the natural example here. Ordinary Lanczos cannot simply be inserted at that point while retaining its symmetric theory.

Eigenvalues alone also miss nonnormal sensitivity. For

$$A_K = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad A_K^{-1} = \begin{bmatrix} 1 & -K \\ 0 & 1 \end{bmatrix},$$

both eigenvalues equal one, but a residual $(0, \epsilon)^{\mathsf{T}}$ corresponds to correction $(-K\epsilon, \epsilon)^{\mathsf{T}}$. Its norm is $|\epsilon|\sqrt{K^2+1}$. A residual tolerance selected from the two eigenvalues would miss the amplification by K. Singular values and the chosen physical norm address this issue directly.

4.5 An exact two-step GMRES check

A small nonsymmetric example exposes every part of the Krylov calculation. Choose

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad s_0 = 0.$$

The first basis vector is $v_1 = (1,1)^{\mathsf{T}}/\sqrt{2}$. Multiplication gives $Av_1 = (3,1)^{\mathsf{T}}/\sqrt{2}$, so $h_{11} = v_1^{\mathsf{T}} A v_1 = 2$. Removing this component leaves $v_2 = (1,-1)^{\mathsf{T}}/\sqrt{2}$, with $h_{21} = 1$. The first GMRES least-squares problem is therefore

$$\min_y \left\| \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} y \right\|_2 .$$

Its minimizer is $y = 2\sqrt{2}/5$, producing correction $s_1 = (2/5, 2/5)^{\mathsf{T}}$ and residual $b - As_1 = (-1/5, 3/5)^{\mathsf{T}}$. The residual norm is $\sqrt{2/5}$, a factor $1/\sqrt{5}$ smaller than the initial norm $\sqrt{2}$. This is the actual residual reduction, rather than a prediction from an eigenvalue plot.

For the second product, $Av_2 = v_2$. Thus the projected two-dimensional operator is

$$H_2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \qquad y_2 = (1/\sqrt{2}, -1/\sqrt{2})^{\mathsf{T}}.$$

It satisfies $H_2 y_2 = (\sqrt{2}, 0)^{\mathsf{T}}$, and $[v_1\ v_2] y_2 = (0,1)^{\mathsf{T}}$ solves the original equation exactly. There is no third independent vector: the Krylov space already fills the two-dimensional state space. Roundoff and inexact products would turn this exact termination into a finite residual check. This example is also an appropriate regression calculation for a GMRES implementation: the basis, projected matrix, first correction, and final answer are all known without a tolerance-dependent reference run.

The same two-dimensional arithmetic can describe scaled spring coefficients, temperatures, or a selected tangent block from a flow mesh. Its exact termination supplies no criterion for choosing among those physical readings. That choice is carried by the original equations, units, and map from coefficients to an observable. This is why a Lanczos or Arnoldi recurrence cannot identify Navier–Stokes or quantum electrodynamics from its final loop alone.

4.6 Why multiple grid scales help

Consider the dimensionless one-dimensional Dirichlet Laplacian

$$(A_h e)_j = (2e_j - e_{j-1} - e_{j+1})/h^2, \qquad e_0 = e_{n+1} = 0.$$

Weighted Jacobi updates an error by $e^{\text{new}} = (I - \omega D_h^{-1} A_h)e$, where $D_h = 2I/h^2$. On the sine mode $e_j = \sin(j\theta)$,

$$e_j^{\text{new}} = \mu(\theta) e_j, \qquad \mu(\theta) = 1 - \omega(1 - \cos\theta). \tag{4.7}$$

For the Dirichlet problem $\theta = k\pi/(n+1)$. Choosing $\omega = 2/3$ gives $|\mu(\theta)| \le 1/3$ for the high-frequency range $\pi/2 \le \theta \le \pi$. A sweep reduces those mode amplitudes by at least a factor of three. For low frequency, $\mu(\theta) = 1 - \theta^2/3 + O(\theta^4)$, so the longest waves hardly change on a fine grid. Smoothing is effective at removing rapid oscillations, not at transporting the longest-scale correction through the whole domain.

A coarse grid represents those long waves with fewer unknowns. Let P interpolate a coarse correction to the fine grid, let R restrict a fine residual, and define the Galerkin coarse operator $A_H = RA_hP$. One coarse correction acts on the error as

$$e^{\text{new}} = (I - PA_H^{-1}RA_h)e. \tag{4.8}$$

If $e = Pe_H$, the error is annihilated exactly when the coarse solve is exact. A two-grid cycle with pre- and post-smoothing has error matrix

$$E_{2g} = S_{\text{post}}(I - PA_H^{-1}RA_h)S_{\text{pre}}.$$

Recursively approximating the coarse inverse gives a multigrid V-cycle. Each level performs a stated number of smoothing sweeps, restricts the remaining residual, receives an interpolated correction, and smooths again. The coarsest system needs a declared solution tolerance; the word multigrid alone specifies none of these operators.

For symmetric positive-definite A_h with $R = P^{\mathsf{T}}$, the coarse correction is an energy-orthogonal projection. Indeed its remainder $e' = e - Pc$, with $c = (P^{\mathsf{T}}A_hP)^{-1}P^{\mathsf{T}}A_he$, satisfies $P^{\mathsf{T}}A_he' = 0$. Therefore

$$e^{\mathsf{T}}A_he = (e')^{\mathsf{T}}A_he' + (Pc)^{\mathsf{T}}A_h(Pc).$$

The correction cannot increase this energy norm. A full mesh-independent convergence bound still requires compatible smoothing and approximation estimates. Convection, anisotropy, strong coefficient contrasts, or indefiniteness can require different smoothers, transfer operators, and coarse spaces [1].

A V-cycle can have linear arithmetic cost under stated mesh assumptions. With a bounded number of stencil operations and smoothing sweeps per unknown, a one-dimensional hierarchy with $n_\ell \le n/2^\ell$ unknowns costs a constant times $\sum_{\ell \ge 0} n/2^\ell \le 2n$, plus its bottom solve. Uniform full coarsening in

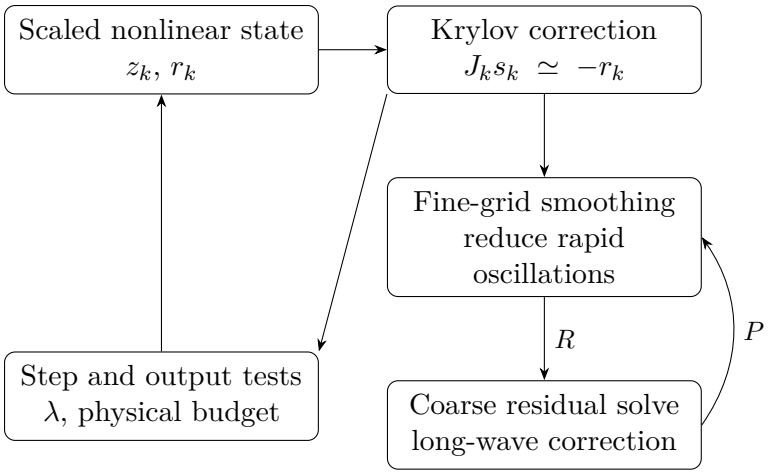


Figure 4.1: Newton controls nonlinear curvature; Krylov methods approximate the tangent solve; multigrid approximates an inverse over several spatial scales. The output budget remains an external acceptance requirement.

three dimensions gives the analogous geometric factor $\sum_{\ell \geq 0} 8^{-\ell} = 8/7$. These counts describe one cycle, not the number of cycles needed for a requested physical tolerance. Dense coarse operators, costly constitutive evaluations, communication, or a bottom solve whose size grows with the fine problem can change the cost model.

Anisotropy illustrates why a smoother is a physical choice. In a diffusion operator $-\partial_x(a_x \partial_x u) - \partial_y(a_y \partial_y u)$, a large ratio a_x/a_y couples neighboring coefficients much more strongly along one direction. Pointwise Jacobi may then fail to reduce an error that is slowly varying in the strongly coupled direction but rapidly varying in the weak direction. Solving whole lines or planes, semi-coarsening, or using an operator-aware coarse space can restore the missing response. The appropriate calculation depends on the coefficients and boundary conditions; the isotropic one-dimensional factor in (4.7) is not a universal smoothing certificate.

4.7 An exact three-unknown coarse-grid calculation

All the preceding operations can be checked without floating point on a small matrix. Suppress the common h^{-2} factor and take

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \qquad P = \begin{bmatrix} 1/2 \\ 1 \\ 1/2 \end{bmatrix}, \qquad R = P^{\mathsf{T}}.$$

Direct multiplication gives

$$AP = (0, 1, 0)^{\mathsf{T}}, \qquad A_H = RAP = 1.$$

For $b = (0, 1, 0)^{\mathsf{T}}$ and initial approximation zero, the restricted residual is $Rb = 1$. The one-unknown coarse solve gives correction one, and interpolation gives $u = P$. Since $AP = b$, this correction solves the fine problem exactly. The result is special to this right-hand side, which lies in the coarse represented response; it is not a claim that one coarse unknown solves every three-unknown load.

The highest sine mode has vector $v = (\sqrt{2}/2, -1, \sqrt{2}/2)^{\mathsf{T}}$ and eigenvalue $2 + \sqrt{2}$. For $\omega = 2/3$, one weighted Jacobi sweep multiplies it by $(1 - \sqrt{2})/3$, whose absolute value is below $1/3$. A represented smooth error is removed by coarse correction, while this rapidly alternating mode is strongly reduced by smoothing. These two exact calculations explain the separate jobs in the V-cycle. Their combination must still be checked for the actual discretized operator and its boundary conditions.

For the flow tangent, a physics-based block preconditioner commonly approximates both the momentum response and the pressure Schur complement from (3.5). A velocity-only diffusion multigrid solve does not by itself enforce continuity. Constraint-compatible coarse spaces and the pressure reference remain necessary. The Schrödinger operator, a gauge-fixed field equation, and the first-order gravitational system have different signs, fields, and null spaces; the same sequence of solver names does not identify their differential equations.

4.8 A finite solution and acceptance procedure

The equations above define a practical nested computation. Its input is a state with admissible material parameters and boundary data, a selected physical output, and separate allowances for algebraic, discretization, data, and reporting errors. The sequence of operations is:

1. Assemble and scale the momentum and constraint residuals. Record both the combined norm and separate continuity or gauge norms. Remove a known pressure null mode by the declared reference condition.

2. Choose a forcing fraction η_k above the estimated tangent and residual evaluation floor. Build or update the approximate momentum and Schur responses used by the preconditioner. A stale preconditioner can be retained if the measured linear convergence remains adequate.

3. Apply right-preconditioned GMRES to the tangent equation, checking the untransformed residual needed in (4.5). Bound any finite-difference product error in that check. Return an incomplete-solve status when the allowed work is exhausted; a fixed iteration count alone is not convergence.

4. Test the proposed state and the line-search decrease. Reduce the step when required by admissibility or nonlinear curvature. After an accepted update, reevaluate the full nonlinear residual rather than reusing the final linear Krylov residual as its replacement.

5. Convert the final residual and constraint allowances to the selected output allowance using established sensitivity bounds. Check mesh, time, input, and reporting contributions separately before declaring the physical output accepted.

A small update is not an alternative stopping proof. In a badly scaled or nearly singular problem the step can become small because arithmetic has stalled, because the line search repeatedly rejects motion, or because the preconditioner has hidden an important component. Conversely a large update in an unobserved null direction can leave a selected physical reading unchanged. Monitoring the actual balance, constraints, and output separates these cases.

For a linear output $q = c^{\mathsf{T}} z$ of a fixed invertible tangent, solving $J^{\mathsf{T}} \lambda = c$ gives exactly $q(z) - q(z_*) = \lambda^{\mathsf{T}} r$ for the associated linear equation. Then $\|\lambda\|_2 \, \|r\|_2$ is a direct output allowance. If a $0.02\,\mathrm{Pa}$ algebraic allocation is available and the calibrated scaled sensitivity is at most $4\,\mathrm{Pa}$ per residual unit,

a sufficient residual threshold is 0.005. A nonlinear use needs an additional remainder allowance as in the preceding chapter. This makes the stopping criterion a statement about the intended pressure reading, rather than an arbitrary number of digits in a solver log.

4.9 Verification by a prescribed exact flow

A reproducible verification problem separates solution error from an unknown experimental answer. For this benchmark all positions, time, velocity, pressure, viscosity, and force are dimensionless, obtained using fixed length, speed, advective-time, and density scales. On a 2π-periodic square, choose unit density, dimensionless viscosity $\nu > 0$, pressure zero, and stream function $\psi = e^{-t} \sin x \sin y$. Then

$$u = e^{-t}(\sin x \cos y, -\cos x \sin y), \qquad \nabla \cdot u = 0, \quad u_t = -u, \quad \Delta u = -2u.$$

Differentiation gives $(u \cdot \nabla)u = e^{-2t}(\sin x \cos x, \sin y \cos y)$. Consequently the exact forcing

$$f = (-1 + 2\nu)u + e^{-2t}(\sin x \cos x, \sin y \cos y) \tag{4.9}$$

makes the momentum equation hold pointwise. All derivatives, initial values, and periodic boundaries are known. This tests the implemented balance rather than fitting its force to computed errors.

For this exact exponential time dependence, the backward-Euler temporal consistency defect is

$$\left[\frac{1 - e^{\Delta t}}{\Delta t} + 1\right] u(t).$$

Since $0 \le e^a - 1 - a \le a^2 e^a/2$ for $a \ge 0$, its norm is at most $\Delta t\, e^{\Delta t} \|u(t)\| /2$. This is a local residual defect, not yet a global state-error estimate; stability and accumulation over time steps supply that conversion. It provides a concrete first-order quantity to check when the spatial error has been made smaller.

Compute the selected velocity and pressure norms on meshes $h, h/2, h/4$ and time steps small enough to isolate spatial error, then repeat with fixed fine space and time steps $\Delta t, \Delta t/2, \Delta t/4$. For exact-solution errors $E_h, E_{h/2} > 0$, report $p_{\text{obs}} = \log_2(E_h/E_{h/2})$. A value near the method's expected order is evidence that the tested range has reached its asymptotic behavior. It is not an enclosure for another problem. Include the constraint norm and the selected output error, because a small velocity RMS error can hide a poor pressure output or a local conservation defect.

Finally lower the nonlinear tolerance by a factor of ten while keeping mesh, time step, and inputs fixed. If the output change is much smaller than its allocated algebraic allowance, further nonlinear work is unlikely to improve the intended reading. If it is larger, tighten the solve or revisit the residual-to-output sensitivity estimate. Then refine the mesh and time step separately and compare with their allocated allowances. For a 0.10 Pa total deterministic pressure-output allowance, an example allocation is 0.02 to the algebraic solve, 0.03 to spatial and temporal approximation, 0.02 to calibrated input data, 0.02 to model discrepancy, and 0.01 Pa to evaluation and reporting. Each term needs its own bound or supporting measurement. Summing these allowances is transparent; running a Krylov loop below machine precision would supply none of the missing physical information.

Chapter 5

Verification against physical field equations

A tolerance becomes an acceptance criterion only after the physical quantity, units, boundary data, and numerical representation have been fixed. A gravity calculation may report tidal curvature; a wavefunction calculation may report a detector probability; a fluid calculation may report velocity or dissipated energy; an electromagnetic calculation may report electric field or radiated energy. The same dimensionless threshold cannot be attached to these readings until their dimensional scales and error propagation have been specified.

This chapter supplies test problems and conditional estimates for the four action presentations. They test equations, constraints, phase, dissipation, and readout accuracy. The fourth-variation identity is an additional check on assembling an action in its declared independent fields. It does not replace these tests. The distinction between verification of a discretization and validation against physical data is standard in engineering; manufactured solutions provide a particularly useful verification technique [18].

5.1 The numerical experiment and its record

Record the dimensional constants and reference scales, initial data, spatial and time boundaries, gauge or coordinate conditions, mesh or spectral cutoff, quadrature, time integrator, nonlinear tolerance, and linear stopping criterion. Specify whether a residual is measured at collocation points, in a quadrature-weighted norm, or as the defect of a continuous reconstruction. A vanishing residual at finitely many points need not bound the defect between those points. Fixed discretization means that the field interpolation and quadrature are fixed while the nonlinear iteration runs; changing the mesh begins a new comparison.

A useful normalized spatial norm is the root-mean-square norm

$$\|w\|_{\mathrm{rms}} = \left(\frac{1}{|\Omega|} \int_{\Omega} |w(x)|^2 \, \mathrm{d}x \right)^{1/2} .$$

Its units are the units of the field. For a probability-normalized wavefunction we instead use its ordinary L^2 norm, because its integral already has a specified physical normalization. Numerical quadrature must approximate whichever integral has actually been selected. The weights cannot be

silently changed when comparing two resolutions.

For an exact reference solution, report the measured numerical error in that norm. For a general solution, report the estimator and the assumptions that make it an upper bound. A mesh-refinement ratio is evidence about observed discretization order; it is not an unconditional bound on the finest-grid error. Input uncertainty, unresolved boundary flux, and model discrepancy require their own allowances. The examples below are analytical benchmarks and computed analytical bounds, rather than reported runs of a production PDE code.

5.2 Gravity: a nonconstant flat-coordinate benchmark

For the first-order gravitational presentation, assemble both

$$\mathcal{R}_{\mu\nu} = R_{(\mu\nu)}(\Gamma), \qquad \mathcal{C}_\lambda{}^{\mu\nu} = \nabla_\lambda h^{\mu\nu}.$$

On a length scale L, compare $L^2\mathcal{R}$ and $L\mathcal{C}$ in fixed positive component norms. These are numerical norms in a declared coordinate frame; contracting with the Lorentzian metric is not a positive error norm. Monitor nondegeneracy of the reconstructed metric independently, together with the chosen coordinate condition and boundary residuals. A small Ricci residual without compatibility need not describe the intended metric connection.

A useful exact test has nonconstant fields but zero curvature. Starting from Minkowski coordinates, replace the spatial coordinate by

$$X = x + \frac{a}{k}\sin(kx), \qquad Y = y, \quad Z = z, \quad X^0 = x^0, \qquad J(x) = 1 + a\cos(kx), \quad |a| < 1.$$

Take x periodic with period $2\pi/k$. The Jacobian J is positive, so this is a smooth periodic coordinate change. The metric and its densitized inverse are

$$g_{\mu\nu} = \mathrm{diag}(1, -J^2, -1, -1), \qquad h^{\mu\nu} = \mathrm{diag}(J, -J^{-1}, -J, -J).$$

The only nonzero connection coefficient is $\Gamma^x_{xx} = J'/J$. Every Ricci component vanishes: in the only potentially nontrivial component, the two derivative terms cancel and the two connection-product terms cancel. Metric compatibility also holds. For instance,

$$\begin{aligned}
\nabla_x h^{00} &= J' - \frac{J'}{J}J = 0, \\
\nabla_x h^{xx} &= \frac{J'}{J^2} + 2\frac{J'}{J}\left(-\frac{1}{J}\right) - \frac{J'}{J}\left(-\frac{1}{J}\right) = 0.
\end{aligned}$$

The final terms in these expressions include the weight-one density connection term, a detail that this benchmark explicitly tests.

Choosing $a = 1/4$ gives $3/4 \le J \le 5/4$ and $9/16 \le J^2 \le 25/16$. Also $|\Gamma^x_{xx}| \le k/3$ follows directly from $|J'| \le k/4$. Thus the test has finite, declared field and conditioning scales. Curvature in every observer frame is exactly zero. It tests derivatives, connection products, density weights, and coordinate handling without introducing matter or a gravitational-wave approximation. To use it as an evolution benchmark, prescribe coordinate source functions and initial/boundary data consistent with this chart; it is not a free test of an incompatible coordinate driver.

For a general gravitational calculation, a residual-to-observable bound still needs the gauge-fixed evolution or boundary operator's stability estimate. Coordinate freedom would otherwise supply

zero modes. A finite matrix inverse bound at one resolution controls that matrix; mesh-independent convergence additionally requires consistency and a uniform bound in the stated continuum norm. Neither the flat benchmark nor the cubic action establishes such a bound for every vacuum spacetime.

5.3 Schrödinger: phase, spectral truncation, and detector error

For a continuous numerical reconstruction $\psi_h(t)$ in the Hamiltonian domain, define the dimensional defect

$$d_h = i\hbar\partial_t\psi_h - H\psi_h.$$

Self-adjoint evolution and variation of constants give the explicit bound

$$\|\psi_h(T) - \psi(T)\|_{L^2} \le \|\psi_h(0) - \psi(0)\|_{L^2} + \frac{1}{\hbar}\int_0^T \|d_h(t)\|_{L^2}\,\mathrm{d}t. \tag{5.1}$$

Indeed, multiplying the error equation by the inverse unitary propagator leaves the integral of $d_h/(i\hbar)$, and unitarity preserves its norm. This estimate has no growing stability factor. Its hypotheses include a common self-adjoint Hamiltonian and compatible boundary data. A piecewise reconstruction with jumps requires the jump magnitudes in the right-hand side as well. A discrete algebraic residual becomes this defect only after reconstruction and consistency errors are included.

For an explicit computed phase test, take a free particle on a periodic interval of length L. Put

$$\phi_n(x) = L^{-1/2}e^{2\pi inx/L}, \qquad \omega_n = \frac{\hbar}{2m}\left(\frac{2\pi n}{L}\right)^2, \qquad \psi(t) = \frac{\phi_0 + e^{-i\omega_1 t}\phi_1}{\sqrt{2}}.$$

The state is normalized. Its relative phase controls interference, so an error in that phase has an observable consequence even if the computed norm is exactly one. A Fourier representation retaining these two modes has zero spatial truncation error for this initial state.

Crank–Nicolson time stepping multiplies the nonconstant coefficient by

$$g = \frac{1 - i\omega_1\Delta t/2}{1 + i\omega_1\Delta t/2} = \exp\left[-2i\arctan(\omega_1\Delta t/2)\right].$$

It preserves amplitude but changes frequency. Let $T = N\Delta t$ be one exact cycle, $\omega_1 T = 2\pi$. The accumulated phase error and state error are exactly

$$\delta_N = 2\pi - 2N\arctan(\pi/N), \qquad e_N = \sqrt{2}\,|\sin(\delta_N/2)|.$$

For $z \ge 0$, differentiating $z - 2\arctan(z/2)$ gives $z^2/(4+z^2) \le z^2/4$. Integration therefore proves

$$0 \le z - 2\arctan(z/2) \le z^3/12, \qquad e_N \le \frac{(2\pi)^3}{12\sqrt{2}\,N^2}.$$

With $N = 400$, the exact bound is approximately $9.135312 \cdot 10^{-5} < 10^{-4}$; the exact error formula gives approximately $9.134974 \cdot 10^{-5}$. These numbers are evaluations of the displayed analytical expressions. They verify the promised state threshold in exact time-stepping arithmetic; floating-point and linear solve errors must fit an additional allowance. For normalized states, the probability error for any detector region is consequently below $2 \cdot 10^{-4}$ using the bounded-observable estimate of Chapter 2.

A general free initial state $\sum_n a_n \phi_n$ adds a transparent spatial budget. Discarding modes outside the retained set I gives exactly

$$\epsilon^2_{\text{space}} = \sum_{n \notin I} |a_n|^2,$$

and this tail norm is constant under exact free evolution. The retained mode phase errors are weighted by their $|a_n|^2$ in the orthogonal L^2 sum. An unrecorded tail is not bounded by showing that retained modes conserve probability. Renormalizing a truncated state also changes the initial error and must be accounted for explicitly.

5.4 Navier–Stokes: an exact viscous-decay experiment

Take the periodic square $(0, 2\pi L)^2$, zero forcing, and define $\xi = x/L$, $\eta = y/L$, and $b(t) = e^{-2\nu t/L^2}$. The fields

$$\begin{aligned} u_x &= Ub(t) \sin\xi \cos\eta, & u_y &= -Ub(t) \cos\xi \sin\eta, \\ p &= \frac{U^2 b(t)^2}{4}(\cos 2\xi + \cos 2\eta) \end{aligned}$$

are an exact Taylor–Green solution with mean-zero kinematic pressure. The divergence vanishes. The Laplacian is $\Delta u = -2u/L^2$, so $\partial_t u = \nu \Delta u$. Direct differentiation gives $(u \cdot \nabla)u = (U^2 b^2/(2L))(\sin 2\xi, \sin 2\eta)$, which is canceled by ∇p. Thus the momentum and incompressibility residuals vanish exactly, not merely to first order in U.

Spatial averaging gives

$$\begin{aligned} \|u\|_{\text{rms}} &= \frac{Ub}{\sqrt{2}}, & K &= \tfrac{1}{2}\|u\|^2_{\text{rms}} = \frac{U^2 b^2}{4}, \\ \nu \|\nabla u\|^2_{\text{rms}} &= \frac{\nu U^2 b^2}{L^2} = -\frac{\mathrm{d}K}{\mathrm{d}t}. \end{aligned} \tag{5.2}$$

At $\nu T/L^2 = 0.1$, the velocity amplitude ratio is $e^{-0.2} \simeq 0.8187307531$ and the energy ratio is $e^{-0.4} \simeq 0.6703200460$. A code must reproduce both the field and its decay to the declared tolerance. The response fields can be set to $v = q = 0$, providing the exact stationary lift of the same solution.

A conditional error estimate can also be derived rather than assumed. Let a smooth reconstructed candidate w be exactly divergence-free and periodic, with momentum defect d and the same viscosity. Put $e = w - u$. Subtracting the two momentum equations, pairing with e, and integrating cancels the pressure and the transport by w. The result is

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|e\|^2_{\text{rms}} + \nu \|\nabla e\|^2_{\text{rms}} \le \|D(u)\|_\infty \|e\|^2_{\text{rms}} + \|d\|_{\text{rms}} \|e\|_{\text{rms}}.$$

Here the matrix norm in $\|D(u)\|_\infty$ is the operator norm of the symmetric velocity gradient. The scalar differential inequality, with a limiting argument at zero error, yields

$$\begin{aligned} \|e(T)\|_{\text{rms}} &\le e^{A(T)} \left[\|e(0)\|_{\text{rms}} + \int_0^T \|d\|_{\text{rms}} \, \mathrm{d}t \right], \\ A(T) &= \int_0^T \|D(u)\|_\infty \, \mathrm{d}t. \end{aligned} \tag{5.3}$$

For this exact flow, $\|D(u)\|_\infty \le Ub/L$, so $A(T) \le \mathrm{Re}(1-b(T))/2$ with $\mathrm{Re} = UL/\nu$. At $\mathrm{Re} = 10$ and $\nu T/L^2 = 0.1$, the amplification bound is approximately $2.475262 < 2.476$. Zero initial error and $\int_0^T \|d\|_{\mathrm{rms}}\,\mathrm{d}t \le 4\cdot 10^{-5}U$ therefore imply $\|e(T)\|_{\mathrm{rms}} < 10^{-4}U$. The residual allowance has units of velocity because an acceleration defect has been integrated over time.

The same velocity bound also controls a specified energy readout. If $K(w) = \|w\|_{\mathrm{rms}}^2/2$, then

$$|K(w) - K(u)| \le \|u\|_{\mathrm{rms}}\,\|w-u\|_{\mathrm{rms}} + \tfrac{1}{2}\,\|w-u\|_{\mathrm{rms}}^2 .$$

This follows by expanding $|u+e|^2$ and bounding the cross term. At the selected time, $\|u\|_{\mathrm{rms}}/U = e^{-0.2}/\sqrt{2}$. The velocity tolerance $10^{-4}U$ therefore implies an energy-per-mass error below $5.790\cdot 10^{-5}U^2$, approximately $3.455\cdot 10^{-4}$ of the exact remaining kinetic energy. For this two-dimensional field, multiplication by $\rho|\Omega|$ gives kinetic energy per unit out-of-plane depth. A uniform layer of thickness H and depth-independent velocity has total energy $\rho H|\Omega|K$. This dimensional conversion illustrates why a velocity tolerance and an energy tolerance cannot be interchanged simply because the same flow generated them.

The assumptions matter. A candidate with a divergence defect requires additional constraint terms or a quantitatively bounded divergence-free reconstruction before this estimate applies. An unmeasured boundary flux invalidates the periodic cancellation. An uncertainty in ν changes the physical decay: a relative viscosity perturbation δ changes the amplitude ratio at this time by $|e^{-0.2\delta} - 1| \le 0.2|\delta|e^{0.2|\delta|}$. For $|\delta| \le 10^{-3}$ this relative amplitude allowance is below $2.001\cdot 10^{-4}$. The allowed perturbation $\delta = -10^{-3}$ produces an rms velocity change approximately $1.158\cdot 10^{-4}U$, which exceeds the preceding solver target. Reducing algebraic residuals cannot remove this input uncertainty. The estimate is for this smooth finite-time benchmark and compatible candidates; it is not a global regularity theorem for every three-dimensional flow.

5.5 QED: electromagnetic constraints and fermion block checks

Separate the bare field-equation verification from a quantum observable calculation. In the first task, record the fermion ordering, coefficient basis or regulator, gauge treatment, spatial domain, and boundary flux. Test the Maxwell residual, both Dirac residuals, Gauss's law, and current continuity. For a finite ordered fermion basis, these are coefficient array tests; the numerical norm refers to that declared basis and its weights. For a quantum observable, also record the quantum state, renormalization prescription, ultraviolet cutoff, finite-volume error, and statistical or quadrature uncertainty. A coefficient residual alone does not bound a renormalized cross section.

An exact coupled-action benchmark is available in the vacuum sector: set $\psi = \bar\psi = 0$ and, in natural units, choose

$$A^0 = 0, \qquad \boldsymbol{A} = (0, A_* \cos(kx - |k|t), 0).$$

It satisfies Lorenz gauge and $\Box A^\mu = 0$. For $k > 0$, $E_y = -kA_*\sin(kx-kt)$ and $B_z = -kA_*\sin(kx-kt)$; all other components vanish. Both divergence constraints are zero, and the mean electromagnetic energy density on a period is $k^2A_*^2/2$. These equalities check phase, relative electric/magnetic amplitude, constraints, and energy. The zero spinors make both Dirac equations and the charge source vanish exactly. This verifies a genuine sector of the displayed action, while leaving its nonzero interaction untested.

There is also an energy estimate for the vacuum Maxwell block. For periodic fields with defects

$\partial_t E - \nabla \times B = r_E$ and $\partial_t B + \nabla \times E = r_B$, subtract an exact solution. Define

$$W(t) = \left(\|e_E(t)\|_{\mathrm{rms}}^2 + \|e_B(t)\|_{\mathrm{rms}}^2 \right)^{1/2},$$
$$r(t) = \left(\|r_E(t)\|_{\mathrm{rms}}^2 + \|r_B(t)\|_{\mathrm{rms}}^2 \right)^{1/2}.$$

Curl integration by parts cancels the cross terms, giving $\frac{1}{2}\partial_t W^2 \le W r$. Integration with a limiting argument at $W = 0$ yields

$$W(T) \le W(0) + \int_0^T r(t)\,\mathrm{d}t.$$

Initial divergence constraints and their propagation must be checked in addition to this evolution estimate. A conductor or radiation boundary adds its actual boundary contribution. Restoring SI units requires the energy weights ϵ_0 and $1/\mu_0$, rather than adding raw electric and magnetic field numbers with different units.

For the free Dirac operator block with prescribed zero potential, $H_D = \boldsymbol{\alpha} \cdot \boldsymbol{k} + \beta m$ on a Fourier mode. The anticommutation relations give $H_D^2 = (|\boldsymbol{k}|^2 + m^2)I$, so the frequencies are $\pm\sqrt{|\boldsymbol{k}|^2 + m^2}$. This provides independent tests of matrix signs, dispersion, and unitary propagation of coefficient spinors. A nonzero charged spinor on a zero potential is not automatically a solution of the coupled Maxwell equation; this is an operator-block test. The coupled interaction additionally requires current and gauge-consistency tests, for example manufactured coefficient fields with analytically calculated source terms. Those sources belong to the verification problem and must be removed when solving the original source-free model.

Before accepting a run, keep a separate allowance for evaluation and reporting. Analytic benchmarks can be evaluated more accurately than the target discretization, but their rounded printed decimal is not the exact reference field. Numerical differentiation of reference data can also contaminate a residual test; the expressions above permit analytic derivatives instead. A check of the action's fourth amplitude response should use the declared independent fields and fixed quadrature. Gauge fixing, field elimination, or changing normalization during that check changes the calculation being tested. The resulting acceptance record should name the physical observable, the norm, and every retained error allowance alongside the final number.

The numerical constants in these analytical examples, together with the beam, Krylov, multigrid, coefficient-sensitivity, and instrument examples, are independently reproduced by `scripts/engineering_examples.py`. Its ledger is `generated/engineering-examples.json`. Finite matrix and coefficient calculations use exact rational arithmetic; exponential acceptance thresholds use bounded Taylor remainders. This checks the displayed examples without reporting an unperformed PDE simulation.

These acceptance criteria use the vocabulary of each field equation: curvature and compatibility, phase and probability, momentum and viscous energy loss, electric/magnetic energy and charge continuity. Each needs its own physical scales and stability estimate. Their common vanishing fourth action response remains an exact consistency check on the selected presentations, while these additional calculations determine what a finite physical reading can claim.

Chapter 6

A worked scalar reading

The scalar fixture makes three questions separately decidable: which root the iteration selects, how fast an exact update approaches it, and which finite decimal an independent interval certifies. All coefficients, input values, and arithmetic conventions are fixed below. The calculation uses ordinary integers and rational numbers. It reproduces the numerical expressions in the source without running the Lean compiler or measuring its elaborator.

6.1 The residual, its branch, and its units

In normalized coordinates the residual and its first two derivatives are

$$r(y) = 13y^2 - 36y + 18, \qquad r'(y) = 26y - 36, \qquad r''(y) = 26.$$

Expansion gives the exact identity

$$(r'(y))^2 = 360 + 52r(y). \tag{6.1}$$

Thus at either root the derivative magnitude is $6\sqrt{10}$. The first sensitivity is $Dr(y)[v] = (26y - 36)v$ for an input increment v; the mixed second sensitivity is $D^2r(y)[v, w] = 26vw$. A constant second derivative here describes the curvature of a scalar response. Identifying it with a physical acceleration would require the corresponding variable, time scale, and response law.

Write the roots as

$$\alpha = \frac{18 + 3\sqrt{10}}{13}, \qquad \beta = \frac{18 - 3\sqrt{10}}{13}, \qquad d = \alpha - \beta = \frac{6\sqrt{10}}{13} > 0.$$

The vertex is $18/13$. Positivity alone does not choose a root, since both roots are positive. We select the increasing branch $y > 18/13$. The conversion to the reported-output coordinate is

$$x = g_x y, \qquad g_x = \frac{324}{5}. \tag{6.2}$$

Both coordinates are dimensionless in this reproduced calculation. The conversion factor g_x is a declared numerical gain, with no identification with the speed of light. Consequently $g_x\alpha \simeq 137.01129055$ and $g_x\beta \simeq 42.43486330$. The output-coordinate residual is

$$q(x) = 325x^2 - 58320x + 1889568 = 324^2 r(5x/324).$$

This equality verifies the coefficient conversion rather than merely noting that two formulas have similar roots. It also shows that Newton and Halley updates commute with this linear coordinate conversion when performed exactly.

The comparison starts at the jar's lower wall $x_0 = 1296/10 = 129.6$, or $y_0 = 2$. This seed is above the vertex and below α. Its first Newton image is $y_1 = 17/8$, giving $x_1 = 137.7$. These two values enclose the selected root. This is the jar-seeded comparison in Episode 12, lines 104–153; it is distinct from the separate pair-valued crank's initial strip reading elsewhere in the device. Retaining the input prevents their iteration counts from being interchanged.

6.2 What the exact updates prove

The two real-arithmetic maps are

$$N(y) = y - \frac{r(y)}{r'(y)} = \frac{13y^2 - 18}{26y - 36}, \tag{6.3}$$

$$H(y) = y - \frac{2r(y)r'(y)}{2(r'(y))^2 - r(y)r''(y)}. \tag{6.4}$$

The second is Halley's update, although the source names its integer implementation `lagrange ReadingScaled`. The second derivative enters a rational correction; simply replacing the residual by its quadratic Taylor polynomial and solving that polynomial is a different operation. Halley's historical method provides context [11]; the convergence orders required here follow from direct identities.

Proposition 6.1 (Exact error laws for this quadratic)**.** *For $e = y - \alpha$, at points where the maps are defined,*

$$N(y) - \alpha = \frac{e^2}{2e + d}, \qquad H(y) - \alpha = \frac{e^3}{3e^2 + 3de + d^2}. \tag{6.5}$$

Proof. Factor $r(y) = 13e(e + d)$ and write $r'(y) = 13(2e + d)$. Substitute in the Newton formula and subtract α. For Halley, cancel the common factor $2 \cdot 13^2$ in its correction. Its remaining denominator is $(2e + d)^2 - e(e + d) = 3e^2 + 3de + d^2$. Subtracting the correction from e leaves exactly e^3 over that denominator. These are algebraic equalities, with no omitted remainder. □

As $e \to 0$, the Newton error divided by e^2 tends to $1/d$, and the Halley error divided by e^3 tends to $1/d^2 = 169/360$. This proves local quadratic and cubic convergence, respectively. The conclusion concerns exact arithmetic near this simple root; it does not assign a universal number of iterations to either method.

The same identities establish the chosen trajectories. On the increasing branch $2e + d > 0$, so Newton's first step from below the root overshoots it. Thereafter $0 < N(y) - \alpha < e$ whenever $e > 0$; the upper sequence decreases, and its only possible limit in this sector is α. Halley's denominator is $3(e + d/2)^2 + d^2/4 > 0$. For $-d/2 < e < 0$, its new error remains negative and has smaller magnitude: subtracting e^2 from that denominator gives $(e + d)(2e + d) > 0$. Its sequence increases from the seed toward the same root. Neither argument crosses the vertex or silently switches to the smaller root.

Error order also has a coordinate scale. If $E = x - g_x\alpha = g_x e$, the output-coordinate identities are

$$E_N = \frac{E^2}{2E + g_x d}, \qquad E_H = \frac{E^3}{3E^2 + 3g_x dE + (g_x d)^2}.$$

Their limiting coefficients are $1/(g_x d)$ and $1/(g_x d)^2$. The orders survive the conversion, while their numerical constants change. This is why an error reported in normalized coordinates must be converted before comparison with a output tolerance. The plot and the printed table both use reported-output coordinates throughout.

For this fixture the second derivative requires no estimation: its constant value follows from the declared coefficients. This makes a second-order response particularly convenient to retain. In a field problem, assembling or applying a second differential can require additional solves, storage, or boundary information. Iteration count and computational work consequently need separate ledgers. The scalar comparison establishes what its two update formulas achieve with one fixed input; the operator costs of the four model presentations must be evaluated on their own stated domains.

6.3 The integer updates actually compared

Now retain the source's finite arithmetic. Let $S = 10^{24}$ and store $X = Sx$ as an integer. Put $A = 325$, $B = 58320$, $C = 1889568$ and

$$F(X) = AX^2 - BXS + CS^2, \quad F_p(X) = 2AX - BS, \quad F_{pp} = 2A.$$

Episode 12, lines 111–117 and 145–153, becomes

$$X^N_{k+1} = \left\lfloor \frac{A(X^N_k)^2 - CS^2}{2AX^N_k - BS} \right\rfloor, \tag{6.6}$$

$$X^H_{k+1} = X^H_k - \left\lfloor \frac{2F(X^H_k)F_p(X^H_k)}{2F_p(X^H_k)^2 - F(X^H_k)F_{pp}} \right\rfloor. \tag{6.7}$$

The Newton numerator is nonnegative on the reproduced orbit, so natural subtraction does not clip it. Both denominators are positive. The signed division in the Halley expression therefore uses Euclidean floor even when its numerator is negative. These facts are checked at every generated step by the document-local script.

Turn	Newton, floor of updated value	Halley, floor of correction
0	`129.600000000000000000000000`	`129.600000000000000000000000`
1	`137.700000000000000000000000`	`136.953191489361702127659575`
2	`137.016233766233766233766233`	`137.011290527013905233309938`
3	`137.011290807319124540443729`	`137.011290548979457087737301`
4	`137.011290548979457793403451`	`137.011290548979457087737301`
5	`137.011290548979457087737300`	`137.011290548979457087737301`

Table 6.1: Exact reproduction at scale 10^{24}, with the seed counted as turn zero. Entries are integers with an inserted decimal point; they are not floating-point approximations.

The two floors occupy different positions. Newton takes the floor of its updated value. Since X is integral, Halley takes the ceiling of its updated value: $X - \lfloor X - SH(X/S) \rfloor = \lceil SH(X/S) \rceil$. Their one-step rounding errors consequently lie in $(-S^{-1}, 0]$ and $[0, S^{-1})$, respectively. Local convergence order alone cannot decide the last stored digit once this arithmetic scale is reached.

All products and differences inside these expressions are exact integers. The reproduction introduces rounding only at the displayed division, rather than rounding the residual, derivative, and quotient

separately. This matters most in the first Halley step: the residual is negative, so flooring its negative correction and then subtracting it rounds the new value upward. Replacing that operation by division truncated toward zero would implement a different machine and would change the table. The script uses Python's integer floor division only after verifying that the divisor is positive.

Halley first reaches its fixed scaled integer at turn three and remains there through turn six; Newton first reaches its fixed integer at turn five. The former is one unit in the last place above the latter. In particular, stabilization at turn three is not a certificate that Halley's displayed final digit is the root's truncation. The source's four-turn budget distinguishes these trajectories: Newton is still approaching its fixed cell at four, while Halley is already stationary. Four is the declared device input for this comparison, not a general bound on root finding or a physical constant inferred by this script.

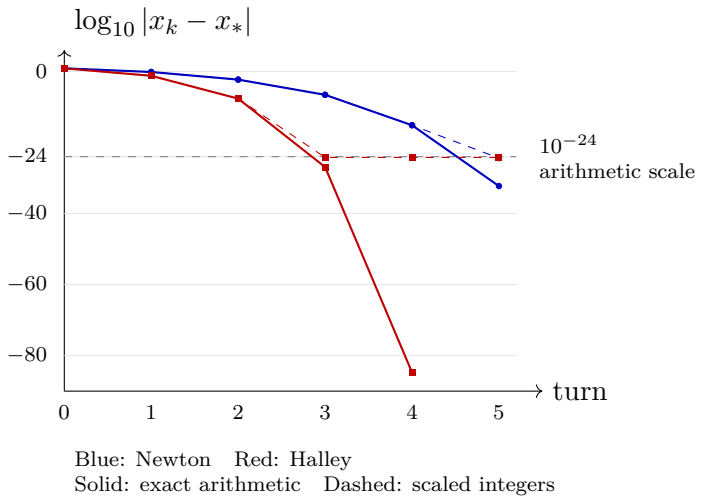


Figure 6.1: Error in reported-output coordinates. Exact-arithmetic curves follow (6.5); dashed curves use the reproduced integer updates. Halley's exact turn-five error is below the plotted range. Decimal calculations with 180-place working precision locate the plotted points; they do not certify the final reporting cell.

6.4 A certificate obtained without either update

An independent bisection starts with $L_0 = 2$, $U_0 = 17/8$. The residual signs are $r(L_0) = -2$ and $r(U_0) = 13/64$. At each step form the exact rational midpoint and replace the wall having its sign. This computation uses neither Newton's derivative nor Halley's second derivative, and takes no iteration output from Table 6.1 as an input.

Because $r' > 0$ throughout the initial interval, continuity and the sign straddle give exactly one enclosed root. Induction preserves that straddle and halves the width. After 108 halvings the normalized bracket endpoints are

$$L_{108} = \frac{548922294368262772954797610095995 7}{259614842926741381426524816461004 8},$$
$$U_{108} = \frac{274461147184131386477398805047997 9}{129807421463370690713262408230502 4}.$$

The exact width after conversion is

$$g_x(U_{108} - L_{108}) = \frac{81}{10 \cdot 2^{108}} < 2.50 \cdot 10^{-32}. \tag{6.8}$$

The 108 steps reproduce the source's separate midpoint budget; they are not the four jar turns. Directed decimal enclosure of the two rational walls yields

$$\begin{aligned} 137.011290548979457087737300387516 95 \ &\le g_x L_{108} < g_x\alpha < g_x U_{108} \\ &\le 137.011290548979457087737300387516 98. \end{aligned}$$

Narrowness by itself would not settle the printed value: an arbitrarily short interval can cross a decimal-cell boundary. The decisive integer checks are

$$\lfloor S g_x L_{108} \rfloor = \lfloor S g_x U_{108} \rfloor = 137011290548979457087737300.$$

These endpoint tests can themselves be stated without decimals. For a normalized bracket endpoint N/D, $D > 0$, and proposed reporting integer K, the required inequalities are

$$5KD \le 324SN < 5(K+1)D.$$

For a residual-sign test, the sign of $r(N/D)$ is exactly the sign of $13N^2 - 36ND + 18D^2$. The midpoint ledger and the reporting test therefore reduce to products, sums, differences, and comparisons of integers. The JSON retains the rational walls as numerator–denominator pairs, so another implementation can check the certificate without trusting its decimal display or its root approximation.

Monotonicity of floor then certifies the 24-place truncation

$$\boxed{137.011290548979457087737300} \tag{6.9}$$

Newton's turn-five integer equals this certified integer. Halley's stationary integer exceeds it by one. Both are excellent approximations, but only the first gives the declared truncation cell. A distinct rounding protocol would need its own cell test.

6.5 Residual size, sensitivity, and reproducibility

For any sign-straddling bracket $[L, U]$ above the vertex, define $m = 26L - 36 > 0$. For a tested point y in that bracket,

$$|r(y)| = \left| \int_\alpha^y r'(t)\,\mathrm{d}t \right| \ge m|y - \alpha|, \qquad |g_x y - g_x \alpha| \le \frac{g_x}{m}|r(y)|.$$

Thus a residual becomes a value-error bound only after retaining a positive derivative margin and the conversion factor. The initial bracket already gives $m = 16$; its refined lower wall gives a stronger bound. Near the vertex the inverse factor would deteriorate. The exact discriminant identity explains why the selected simple root itself has a nonzero response, while the bracket establishes which root is being tested.

The finite numeral remains rational, whereas the selected root is irrational. Indeed, the quadratic discriminant is 360, strictly between 18^2 and 19^2; an integer that is a rational square must be an

integer square after cancelling the fraction. The quadratic formula therefore excludes a rational root. A successful decimal certificate establishes membership in a reporting cell, with a controlled nonzero residual. It preserves the distinction between an exact algebraic value and a usable finite indication.

A small successive change answers another question. Either rounded iteration eventually has zero successive change in its integer state, although its stored value still has nonzero polynomial residual. Its stopping observation certifies a fixed point of the rounded map. The sign-straddling interval, derivative margin, and endpoint-cell inequalities supply the additional statements about the exact root. This example makes the three tests concrete: stable iteration, small residual, and stable displayed digits have different acceptance calculations.

The same derivative margin can carry a model-error contribution. Suppose a perturbed residual $\widetilde{r}$ has a root $\widetilde{\alpha}$ in this bracket and $|\widetilde{r} - r| \le \eta$ there. Then $|r(\widetilde{\alpha})| \le \eta$, and the preceding integral estimate gives $|g_x\widetilde{\alpha} - g_x\alpha| \le g_x\eta/m$. Both the perturbation envelope and preservation of the branch are hypotheses. This is a finite sensitivity calculation that can be added to an instrument's calibrated error budget once its coefficient uncertainties have supplied such an envelope.

6.6 Coefficient uncertainty and calibrated parameter sensitivity

The preceding digit certificate treats every coefficient as exact. A physical identification would additionally have to establish those coefficients, their units, and their uncertainty. To make this distinction quantitative, write the residual as

$$r(y; a, b, d_0) = ay^2 - by + d_0, \qquad (a, b, d_0) = (13, 36, 18).$$

Implicit differentiation at a simple root gives the local sensitivities

$$\frac{\partial y}{\partial a} = -\frac{y^2}{2ay - b}, \qquad \frac{\partial y}{\partial b} = \frac{y}{2ay - b}, \qquad \frac{\partial y}{\partial d_0} = -\frac{1}{2ay - b}.$$

These are derivatives of the root with respect to input coefficients, not extra terms in Newton's correction. They quantify how a calibrated input allowance propagates into the calculated quantity. A standard uncertainty calculation could combine their products with a coefficient covariance matrix. A deterministic acceptance calculation instead needs bounded coefficient intervals and a branch-preservation check [12, 14].

Suppose each coefficient changes in absolute value by at most 10^{-8}. On the complete initial bracket $2 \le y \le 17/8$, the residual perturbation is bounded by

$$\eta \le 10^{-8}\left[(17/8)^2 + 17/8 + 1\right] = \frac{489}{64}\,10^{-8}.$$

This bound is much smaller than either endpoint sign margin: the unperturbed values are $r(2) = -2$ and $r(17/8) = 13/64$. The perturbed residual therefore still changes sign. Its derivative is at least $16 - (2 \cdot 17/8 + 1)10^{-8} > 0$, so it has exactly one root in this bracket. At that perturbed root the original residual has magnitude at most η. The original derivative margin $m = 16$ then yields the finite output allowance

$$|\widetilde{x} - x_*| \le \frac{g_x\eta}{16} = 3.094453125 \times 10^{-7}.$$

This is a rigorous bounded-input example, not a linearized estimate. The assumed 10^{-8} coefficient allowances are illustrative. No experiment in this chapter measures them. Their role is to show how

an explicit parameter uncertainty can dominate a 24-place arithmetic certificate by many orders of magnitude while both calculations remain correct for their stated inputs.

Changing the output gain supplies another contribution. If the physical indication is $Y = gx + b_0$, and its calibrated gain and offset have bounded errors ε_g and ε_b, then a calculated input error $|\delta x| \le \varepsilon_x$ gives

$$|\delta Y| \le |g|\varepsilon_x + |x|\varepsilon_g + \varepsilon_g\varepsilon_x + \varepsilon_b.$$

The gain has the units of Y per unit of x, and the offset has the units of Y. A quantizer adds its own output allowance. This is how a root certificate could enter a length, voltage, or phase measurement. Without such a measurement equation, the number printed here remains a dimensionless result of the specified computation.

6.7 Resolution, stopping, and computational work

The stored scale $S = 10^{24}$ sets a uniform output increment $\Delta_x = 10^{-24}$. Truncation of a positive exact value has one-sided error in $[0, \Delta_x)$; nearest rounding has error at most $\Delta_x/2$ with a declared tie rule. The source's integer Newton and Halley formulas implement different positions of the floor operation. Their fixed integer states consequently differ by one increment even though both approximate the same simple root extremely closely. That difference is completely explained by the arithmetic rules; it is not an observed fluctuation in a physical constant.

For a required output tolerance ε_x, an interval solver can stop when its converted interval has width at most $2\varepsilon_x$ and report its midpoint with that error allowance. A decimal truncation decision needs the stronger same-bin endpoint test already used above. The stopping rules answer different questions: one bounds a numerical error, while the other fixes one particular display code. Repeated displayed digits by themselves supply neither the sign margins nor the interval inclusion.

Newton's exact local error is proportional to the square of the previous error; Halley's is proportional to its cube. Computational cost still depends on the implementation. For this scalar quadratic, $r'' = 26$ is a stored constant, so its evaluation is negligible compared with a general field problem. A multidimensional second response may require additional residual evaluations, derivative storage, tangent solves, or contractions with a large field increment. The total work should therefore be recorded as residual evaluations, linear iterations, preconditioner applications, memory, and wall time for a fixed output tolerance. A comparison using iteration counts alone would omit those different costs.

The 108 bisections supply an independent verification route. They require only exact residual signs at rational points; they use neither Newton's tangent nor Halley's second response. Their work is deliberately redundant with the fast updates. Agreement of the independent bracket and the printed Newton integer is an arithmetic consistency check with a specified branch and rounding convention. It does not identify the calculated dimensionless quantity with a material or electromagnetic measurement without the additional calibration just described.

The reproducible input is `scripts/scalar_reading.py`. Run it from this project to regenerate the table, vector figure, certificate macros, and `generated/scalar-reading.json`; `-check` verifies the retained products. The ledger records every rational bracket and iterate, integer state, coefficient, seed, rounding choice, plotting precision, and inspected source hash. Its assertions check denominator signs, example rows, endpoint truncations, and the stationary integers.

Chapter 7

Calibrated instruments and common readings

A numerical result becomes a physical reading through a measurement chain. The chain starts with a specified specimen, field, or operating condition; a sensor responds to a quantity of interest; signal conditioning applies a gain and an offset; a digitizer supplies a finite record; and a calibration converts that record to the reported units. A simulation has a parallel chain: material and boundary data, a field equation, a discretization, an iterative solution, and an output functional. Agreement at the end of two chains is a statement about those complete chains, including their tolerances. It does not require their internal mechanisms to be identical.

Chapter 6 supplies an unusually precise finite example. The input polynomial, selected root, initial iterate, arithmetic, and reporting scale are specified, so the calculation can be checked down to its printed integer. The four selected actions of Chapter 2 supply equations and their variation hierarchy. This chapter describes the additional calibration needed to compare physical quantities, and identifies exactly what the implemented source computations already check.

7.1 The measurement chain and the quantity being measured

Let s denote the operating state, and let $Q(s)$ denote the scalar or vector measurand, in declared units. For instrument i, write $E_i(s)$ for its ideal internal indication and r_i for its calibration curve. A bound

$$\|r_i(E_i(s)) - Q(s)\| \le \eta_i$$

allows for the discrepancy of the complete idealized instrument from the measurand. This can include probe placement, a sensor model, or a difference between a spatial average and a point value. If the acquired indication $\widehat{z}_i$ satisfies $\|\widehat{z}_i - E_i(s)\| \le \delta_i$, and the calibration has sensitivity bounded by L_i, then

$$\|r_i(\widehat{z}_i) - Q(s)\| \le \eta_i + L_i\delta_i.$$

For a scalar differentiable calibration, one can take $L_i = \sup |r_i'|$ over the admitted indication interval. A steep inverse calibration amplifies small indication errors. A vector calibration uses the induced matrix norm of its sensitivity matrix.

Applying this estimate to two instruments gives

$$\|r_1(\widehat{z}_1) - r_2(\widehat{z}_2)\| \le \eta_1 + \eta_2 + L_1\delta_1 + L_2\delta_2. \tag{7.1}$$

To prove it, add and subtract each calibrated ideal indication and $Q(s)$, then use the triangle inequality. The estimate compares the same operating state and the same measurand. If the probes sample different times or positions, their timing and placement errors belong in η_i; merely using the same unit on both displays does not remove those errors.

An exact common reading has $\eta_i = \delta_i = 0$ for the selected numerical chain. A physical calibration generally carries uncertainty. A worst-case bound such as (7.1) adds admitted error magnitudes. Standard uncertainties, coverage intervals, and probabilistic confidence statements require the additional statistical assumptions described in the measurement model [12, 14]. In particular, a root-sum-of-squares calculation cannot replace this hard bound without a justified treatment of correlations and error distributions.

7.2 A quantitative calibration and digitization budget

Consider two hypothetical displacement instruments observing the same motion x, with $|x| \le 1\,\mathrm{mm}$. Their nominal voltage sensitivities are $a_1 = 100\,\mathrm{V\,m^{-1}}$ and $a_2 = 200\,\mathrm{V\,m^{-1}}$. Within their calibrated range, suppose the actual conditioned voltage obeys

$$v_i = a_i(1 + \gamma_i)x + b_i + \Delta b_i + n_i.$$

Here b_i is the nominal offset, γ_i the fractional gain error, Δb_i the uncorrected offset, and n_i the admitted voltage noise. The reported displacement is $(\widehat{v}_i - b_i)/a_i$, followed by decimal rounding. This model fixes what each error acts on; it is not a claim that all sensors have this linear response.

Take a uniform 12-bit digitizer over a 4 V span. Its ideal code spacing is

$$\Delta v = \frac{4}{4096}\,\mathrm{V} = 0.9765625\,\mathrm{mV}.$$

For nearest-code conversion away from saturation, quantization contributes at most $\Delta v/2$. The corresponding displacement errors are $4.8828125\,\mu\mathrm{m}$ and $2.44140625\,\mu\mathrm{m}$. These values use a declared ideal digitizer: transition-level nonuniformity, reference drift, and amplifier nonlinearity would require their own specifications. A finite code count alone does not bound those additional effects.

Assume $|\gamma_i| \le 2 \times 10^{-4}$, $|\Delta b_i| \le 0.2\,\mathrm{mV}$, $|n_i| \le 0.3\,\mathrm{mV}$, and reporting to the nearest $0.01\,\mu\mathrm{m}$. Let the model and placement errors be $\eta_1 = 1\,\mu\mathrm{m}$ and $\eta_2 = 1.5\,\mu\mathrm{m}$. Direct conversion to displacement gives the following hard budget.

Their readings can differ by at most $17.73421875\,\mu\mathrm{m}$ under these assumptions. A larger disagreement rejects at least one premise of the measurement chain. A smaller disagreement is compatible with the budget; it cannot identify which individual errors occurred. In particular, the instruments may share a reference error or the same mistaken mechanical model. Repeating a measurement can reduce an estimated random component while leaving a common calibration error unchanged.

This example also shows why additional printed digits need not improve a physical tolerance. Reducing the final decimal step by a factor of ten barely changes a budget dominated by sensor noise and digitization. Increasing the analog gain improves displacement resolution only while the full signal, offset, and transient excursions remain within the digitizer range. A complete operating specification therefore declares both a smallest resolvable increment and a largest unsaturated input.

Error contribution	Instrument 1 μm	Instrument 2 μm
Model and placement	1	1.5
Residual offset	2	1
Admitted voltage noise	3	1.5
Ideal code quantization	4.8828125	2.44140625
Gain over the declared motion range	0.2	0.2
Final decimal rounding	0.005	0.005
Maximum total	11.0878125	6.64640625

Table 7.1: An illustrative deterministic displacement budget. The numbers are assumed specifications of the example, rather than measurements of an existing apparatus.

7.3 A common indication and an independent reference

A practical comparison should excite more than one operating point. Zero input tests residual offset; two separated reference displacements test gain; additional points test nonlinearity. Reversing the direction of motion can reveal hysteresis, while a hold at fixed displacement can reveal drift. Each test compares a specified observable over a specified time and bandwidth. A static calibration cannot establish the response to an arbitrarily rapid transient.

There is a simple common-mode failure that any comparison must allow. If two calibrated outputs are both $x + b$, their difference is exactly zero for every x, although both have error b. A shared reference or shared processing model can produce such a correlated bias. An independently established reference constrains that otherwise invisible component. Likewise, running the same polynomial through two storage formats verifies agreement of the processing paths, while measuring an independently prepared specimen tests the additional physical calibration.

For a time-dependent measurement the acquisition specification also declares the sample times and the analog bandwidth. Two digitized records can agree at every sample and differ between samples. Bounding those unobserved excursions requires a stated smoothness, frequency, or dynamics assumption. The associated interpolation tolerance then belongs beside the quantization tolerance, rather than being hidden inside the count of reported digits.

7.4 Exact count channels and storage consistency

The source implements a count channel with no analog uncertainty. For a finite tape t, let $c(t)$ be its number of cells. A computation of the form $F(t) = G(c(t))$ receives only this integer. Replacing every cell's payload while retaining the same number of cells gives $c(t') = c(t)$ and therefore $F(t') = F(t)$. This is an exact input-output invariance of the implemented channel. It is analogous to a pulse counter that records the number of accepted pulses while discarding their individual amplitudes.

Episode 8's selected tape readers count facts. Episode 12's `ComputerProgram.ofTape` constructs a second stored form, and its `read_ofTape` computation preserves the count. The named results `the_other_machine_agrees` and `relettering_is_free` compare the resulting scalar calculation. The former checks another implementation of the reading pair; the latter overwrites cell contents while retaining the count and checks that the same pair is produced. The statement `any_count_of_the_lengths` accepts an alternative counter only when it agrees with tape length on every admitted

tape. Under that explicit premise, it produces the same result.

A detector that reads a cell's payload is a different measurement channel. It may distinguish the two tapes even when the count channel cannot. Similarly, two pulse trains can have the same count while differing in arrival times, energy, spectrum, or polarization. Exact equality of the count is valuable precisely because its scope is fixed. It checks that the two selected implementations deliver the same integer input to the same scalar computation.

Consequently a polynomial coefficient can be inserted explicitly and the two count-based implementations can still agree exactly. The agreement establishes computational consistency for those inputs. Experimental validation would additionally require an independently calibrated link between the coefficients, the apparatus, and the physical measurand. A computed decimal certificate removes arithmetic ambiguity; it does not supply that missing calibration by itself.

7.5 An erased channel and the auxiliary record

Some processing rules deliberately merge all inputs into one reported class. The source contains an exact example. It compares pairs of natural numbers using

$$(a, b)\ \mathcal{R}\ (c, d) \quad \Longleftrightarrow \quad ad \leq cb.$$

For any two pairs, either this comparison or the reverse comparison holds. The implemented class-identification rule treats either direction as sufficient to merge the two records. All pairs are therefore merged. Episode 12 names the resulting statement `the_box_has_one_point`. Any output computed from this single merged class alone is constant.

In measurement language this is a completely erased output channel. The result does not prevent the surrounding apparatus from storing the original numerator, denominator, sign choice, or interval. Those fields constitute an auxiliary record. A familiar example is reporting a squared voltage while separately retaining its polarity bit. The squared channel alone loses the sign; the augmented record can recover it. The separate specification manuscript constructs two different binary auxiliary records and states which measurement each one completes.

The scalar calculation retains a numerator, a positive denominator, the selected upper-root interval, and a bracket certificate. Its decimal reading is computed from those retained data. It is not recovered from the erased channel. Keeping this distinction explicit is part of the instrument specification: every recovered quantity must have a named source in the stored measurements. A calculation cannot reconstruct a discarded distinction unless an additional measurement, prior constraint, or retained record supplies it.

7.6 A finite transition network and the scalar root

A second exact realization uses a two-state transition network. Let

$$M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix}.$$

Interpret M_{ij} as the number of distinct allowed transitions from state i to state j during one clock step. This is a multiplicity matrix, rather than a probability matrix: its rows need not sum to one.

The number of closed transition sequences of length n is $N_n = \operatorname{tr}(M^n)$. Its characteristic polynomial is

$$\chi_M(\lambda) = \lambda^2 - 36\lambda + 234, \qquad \chi_M(\lambda) = 13r(\lambda/13), \quad r(y) = 13y^2 - 36y + 18.$$

The two growth multipliers $\lambda_\pm = 18 \pm 3\sqrt{10}$ therefore give exactly the two roots $y_\pm = \lambda_\pm/13$ used in Chapter 6. The output conversion in that chapter adds its separately declared scale.

Cayley–Hamilton gives

$$N_0 = 2, \qquad N_1 = 36, \qquad N_{n+2} = 36N_{n+1} - 234N_n.$$

For example $N_2 = 828$; both direct matrix multiplication and the recurrence give that integer. Moreover $N_n = \lambda_+^n + \lambda_-^n$, so

$$\lim_{n\to\infty} \frac{\log N_n}{n} = \log \lambda_+.$$

This is a logarithmic transition-count growth rate, in nats per clock step. If one clock step lasts Δt, the corresponding rate per second is $(\log \lambda_+)/\Delta t$. No energy scale, temperature, or probability distribution is specified by these counts. A thermodynamic entropy production rate, measured in $\mathrm{J\,K^{-1}\,s^{-1}}$, is a different calibrated quantity.

The entire sequence also satisfies the generating identity

$$\exp\left(\sum_{n\geq 1} \frac{N_n z^n}{n}\right) = \frac{1}{\det(I - zM)} = \frac{1}{1 - 36z + 234z^2}.$$

For this matrix, expanding $-\log(1-\lambda_+ z) - \log(1-\lambda_- z)$ proves the identity coefficient by coefficient. Its singularities occur at the reciprocals of the multipliers. This specifies the exact finite-network calculation without assigning a physical meaning to an unmeasured clock or energy.

The table entries are construction data. The source itself records that deriving these particular weights from the larger apparatus is an additional obligation. Once they are supplied, the root correspondence is both necessary and sufficient for this specified matrix. Another matrix can share its characteristic polynomial without sharing all its transition counts, spatial arrangement, or physical realization.

7.7 Software instruments, blanks, and independent checks

The elaborator experiments measure computational work under a stated protocol. They retain both a raw sample and a blank, then subtract in signed integer arithmetic. If $H(n)$ is the sample and $B(n)$ the blank, the corrected reading is $C(n) = H(n) - B(n)$; a negative result remains visible. Truncating the subtraction at zero would destroy the evidence of a sample below its blank.

The command `monte_walk` visits the same nesting depths in ascending and descending order before publishing intermediate declarations. Its `thePassGap` compares corrected readings at matching depths. This removes an identified change in the intervening declaration environment. It does not eliminate every source of run-to-run variation. A reproducible performance study would retain software versions, configuration, input fixture, meter definition, warm-up policy, repeated runs, and the corresponding blanks. Heartbeats are the implementation's work units, rather than an SI time or energy measurement.

A second fixture constructs a pure chain and a branched chain of `Variation` constructors. On the pure chain the selected Newton and Krylov traversals have the same output length. On the branched fixture, one retains a single branch and the other retains both. Their specified lengths are $n + 1$ and $1 + n(n + 3)/2$. The measured work difference therefore tests the cost of those implemented traversals.

Interpreting this as a measurement of fluid convection would require a further engineering model: which stored quantity represents velocity, which operation represents spatial differentiation, which clock represents time, and how the measured work relates to force, power, or error. The actual Newton linearization of $(u \cdot \nabla)u$ already contains $(u \cdot \nabla)h + (h \cdot \nabla)u$. A difference in traversal cost cannot replace a check of those physical terms.

Finally, an iterative solver does not uniquely identify the field equation that produced it. If an operator has independent blocks $A = T \oplus C$ and the initial vector excites only T, every observed Krylov vector remains in that block. Changing C changes no observed recurrence. Lanczos additionally assumes an appropriate self-adjoint problem and inner product; convection Jacobians are generally nonsymmetric. Arnoldi/GMRES and an explicitly specified preconditioner preserve that distinction [15, 21]. Multigrid also requires its coarse operators, transfers, smoothers, and boundary treatment.

Numerical verification checks that the implemented calculation solves its declared mathematical problem to its declared tolerance. Instrument validation checks that this problem and its calibration describe the physical measurement in the stated operating range. The scalar bracket and exact count comparisons complete the former finite calculations. Equation (7.1) states the additional error budget through which independent calibrated instruments can test the latter.

Epilogue: Universe numbers only increase

A mechanical cycle can close its displacement reading while retaining a new event. Comparing that count with entropy requires a clock, energy, temperature, and an explicit calibration. The beam model supplies these quantities through a force balance and a thermal measurement equation.

A physical beam and its dissipative equation

Let $x \in [0, L]$ measure position along a slender beam, and let $q(x, t)$ be its small transverse displacement. Both x and q are measured in metres. For constant Young's modulus E and second moment of cross-sectional area I, the flexural rigidity EI has units $\mathrm{N\,m^2}$. The small-slope Euler–Bernoulli bending energy is

$$U[q] = \frac{EI}{2} \int_0^L |q_{xx}|^2 \, \mathrm{d}x. \tag{E.1}$$

Since curvature q_{xx} has units $\mathrm{m^{-1}}$, this expression has units $\mathrm{N\,m} = \mathrm{J}$. The engineering assumptions include a linear elastic bending law and negligible shear deformation in the selected regime [22]. They are part of the material model.

Suppose distributed viscous drag opposes the transverse velocity with force per length $-\zeta q_t$, where $\zeta > 0$ is constant and has units $\mathrm{N\,s\,m^{-2}}$. In the overdamped approximation, force balance is

$$\zeta q_t + EI q_{xxxx} = f(x, t), \qquad [f] = \mathrm{N\,m^{-1}}. \tag{E.2}$$

Each term is force per length. Inertial bending dynamics would also contain μq_{tt}, where μ is mass per length. Omitting that term requires the time scale of interest to make inertia small relative to drag; it is an additional approximation, rather than a consequence of stationarity. The constant mobility of a dimensionless bending formulation corresponds here to EI/ζ, with units $\mathrm{m^4\,s^{-1}}$.

Take $f = 0$ and fixed endpoint displacements and slopes. Their equilibrium q_* is the cubic of Proposition 1.3. The deviation $v = q - q_*$ satisfies homogeneous clamps $v = v_x = 0$ at both ends, and $\zeta v_t + EI v_{xxxx} = 0$. The equilibrium equation is $q_*^{(4)} = 0$. The evolution equation specifies how that equilibrium is approached; the stationary equation alone specifies neither a clock nor a drag law.

Power, boundary work, and the decay estimate

For a sufficiently smooth displacement, differentiating the bending energy and integrating twice by parts gives the full mechanical power balance

$$\begin{aligned}\dot{U} &= \left[EIq_{xx}q_{xt} - EIq_{xxx}q_t\right]_0^L + \int_0^L EIq_{xxxx}q_t \,\mathrm{d}x \\ &= P_{\text{supports}} + \int_0^L f q_t \,\mathrm{d}x - \zeta \int_0^L |q_t|^2 \,\mathrm{d}x. \end{aligned} \tag{E.3}$$

The first bracket is power delivered through the ends with the displayed sign convention. Its moment multiplies endpoint angular velocity, and its shear force multiplies endpoint transverse velocity. Fixed clamps make both endpoint velocities zero. Applied distributed loads supply the second work term. The last term is nonnegative viscous loss, in watts.

With fixed clamps and $f = 0$, equation (E.3) becomes

$$\dot{U} = -\zeta \int_0^L |q_t|^2 \,\mathrm{d}x = -\frac{(EI)^2}{\zeta} \int_0^L |q_{xxxx}|^2 \,\mathrm{d}x \le 0. \tag{E.4}$$

For a nonzero time-independent conservative load, the decreasing quantity is instead the total potential $\Pi = U - \int_0^L fq \,\mathrm{d}x$. A time-dependent load adds $-\int f_t q \,\mathrm{d}x$ to $\dot{\Pi}$. These distinctions matter when interpreting a driven cycle: a decrease in elastic energy and an increase in dissipated heat need not be equal if the supports or actuator supply work.

For the unforced relaxation define the physical energy gap $G(t) = U[q(t)] - U[q_*] = (EI/2)\,\|v_{xx}\|_2^2$. Homogeneous clamps give the elementary estimates

$$\|v\|_2 \le (L/\pi)^2 \,\|v_{xx}\|_2 \,, \qquad \|v_{xxxx}\|_2 \ge (\pi/L)^2 \,\|v_{xx}\|_2 \,.$$

The first follows by applying the interval Poincare inequality to v and then to v_x, whose mean is zero. For the second, integrate to obtain $\|v_{xx}\|_2^2 = \int v_{xxxx} v \,\mathrm{d}x$ and apply Cauchy–Schwarz. Substitution in (E.4) proves

$$G'(t) \le -\kappa G(t), \qquad G(t) \le G(0)e^{-\kappa t}, \qquad \kappa = \frac{2EI}{\zeta}\left(\frac{\pi}{L}\right)^4. \tag{E.5}$$

The exponent is dimensionless because κ has units s^{-1}. This gives a directly usable settling-time estimate: $G(t) \le \varepsilon_G$ is guaranteed once $t \ge \kappa^{-1} \log(G(0)/\varepsilon_G)$ when $0 < \varepsilon_G < G(0)$.

A mode calculation with declared units

The clamped beam has bending modes φ_j with positive stiffness eigenvalues λ_j. Finite integrated initial curvature permits the energy-based evolution [7]. In modal coordinates,

$$v(x,t) = \sum_j a_j \exp\left(-\frac{EI\lambda_j}{\zeta} t\right) \varphi_j(x).$$

The modal solution smooths the displacement at positive times and justifies the energy identities for finite initial bending energy.

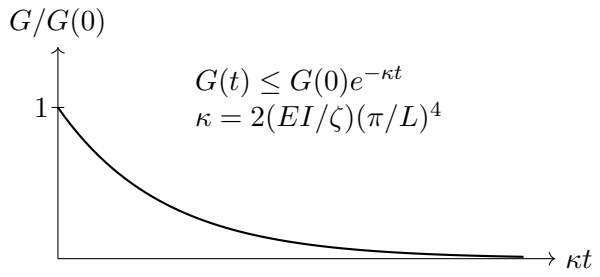


Figure E.1: The energy gap decreases under a specified drag law and fixed clamps. The exponential estimate sets a physical settling-time tolerance. The plotted curve is the normalized exponential upper bound.

For an explicit single-mode example, choose dimensionless φ_1 with $\int_0^L \varphi_1^2 \,\mathrm{d}x = L$. Solving $\varphi_1^{(4)} = \lambda_1 \varphi_1$ with the four clamp conditions gives $\lambda_1 = \beta_1^4/L^4$, where the first positive bending root satisfies $\cosh\beta_1 \cos\beta_1 = 1$ and $\beta_1 \approx 4.73004$. The displacement $v = a\varphi_1 e^{-t/\tau}$ then has

$$\tau = \frac{\zeta L^4}{EI\beta_1^4}, \qquad G(t) = \frac{EI\, a^2 \beta_1^4}{2L^3} e^{-2t/\tau}.$$

The displacement amplitude decays with time constant τ, while its quadratic bending energy decays twice as fast in the exponent.

As an illustrative set of model parameters, take $L = 1\,\mathrm{m}$, $EI = 1\,\mathrm{N\,m^2}$, $\zeta = 100\,\mathrm{N\,s\,m^{-2}}$, and $a = 1\,\mathrm{mm}$. Then $\tau \approx 0.200\,\mathrm{s}$ and $G(0) \approx 2.50 \times 10^{-4}\,\mathrm{J}$. These numbers specify a model calculation, rather than characterize an unmeasured material. If mass is retained, the modal inertial-to-drag ratio on this relaxation scale is $\mu EI\lambda_1/\zeta^2$; the overdamped approximation requires that ratio to be small.

What a time discretization dissipates

A computed relaxation also needs a discrete energy check. In a conforming beam approximation let $\mathbf{v}^n$ contain the displacement coefficients at time $n\Delta t$, with homogeneous clamps enforced. Let $\mathsf{M}_{ij} = \int N_i N_j \,\mathrm{d}x$ be its mass matrix and $\mathsf{K}_{ij} = \int N_i'' N_j'' \,\mathrm{d}x$ its curvature matrix. Backward Euler gives

$$\frac{\zeta}{\Delta t}\mathsf{M}(\mathbf{v}^{n+1} - \mathbf{v}^n) + EI\,\mathsf{K}\mathbf{v}^{n+1} = 0.$$

Set $\mathbf{d} = \mathbf{v}^{n+1} - \mathbf{v}^n$ and $G_h^n = (EI/2)(\mathbf{v}^n)^T \mathsf{K}\mathbf{v}^n$. Multiplication by $\mathbf{d}^T$ yields the exact identity

$$G_h^{n+1} - G_h^n = -\frac{\zeta}{\Delta t}\mathbf{d}^T\mathsf{M}\mathbf{d} - \frac{EI}{2}\mathbf{d}^T\mathsf{K}\mathbf{d} \le 0.$$

Thus the fully solved step cannot increase this discrete bending energy for any positive time step. The first loss term is the time-step integral of the resolved viscous power. The second is additional numerical damping from backward Euler.

Assigning all discrete energy loss to the physical reservoir would also assign the algorithm's extra damping to it. Spatial and time refinement must control this error before the result supplies a physical entropy reading. Inexact linear solves additionally contribute residual work to the same energy budget.

Heat transfer and an entropy calibration

Assume the viscous loss is delivered as heat to a large reservoir at fixed absolute temperature $T > 0$, with no applied work in the selected relaxation. The reservoir then receives the power $-\dot{U}$, and

$$\dot{S}_{\text{reservoir}} = -\frac{\dot{U}}{T} = \frac{\zeta}{T}\int_0^L |q_t|^2 \, \mathrm{d}x = \frac{(EI)^2}{\zeta T}\int_0^L |q_{xxxx}|^2 \, \mathrm{d}x \geq 0. \tag{E.6}$$

Entropy is measured in $\mathrm{J\,K^{-1}}$ and its rate in $\mathrm{W\,K^{-1}}$. This equation calculates the reservoir entropy change. A statement about total entropy production must also account for the beam's thermal state, the drag medium, and any other energy stores. An isothermal free-energy model can provide that complete account. The second law permits zero total production in the reversible limit [9, 17].

The entropy transferred during a time interval is

$$\Delta S = \frac{\zeta}{T}\int_{t_0}^{t_1}\int_0^L |q_t|^2 \, \mathrm{d}x \, \mathrm{d}t = \frac{U[q(t_0)] - U[q(t_1)]}{T}.$$

At equilibrium, $q^{(4)} = 0$, the drag power vanishes, and the cubic minimizes bending energy for its fixed clamps. Thus the variational cubic is the equilibrium of this explicit dissipative realization. The energy balance, drag coefficient, and reservoir specify the direction of relaxation. The spatial stationary equation alone supplies none of those thermal premises.

At $T = 300\,\mathrm{K}$, the single-mode example can transfer at most about $8.34 \times 10^{-7}\,\mathrm{J\,K^{-1}}$ to the reservoir. This finite capacity is a useful engineering limit: the initially stored elastic energy cannot supply indefinite heat production. A subsequent cycle requires an actuator or another energy source to restore the displacement, and its work and heat must enter the complete balance.

A threshold reader and its retained count

Let $\sigma > 0$ be the entropy increment assigned to one registered pulse. For the unforced beam define

$$\Sigma(t) = \frac{U[q(0)] - U[q(t)]}{T}, \qquad g_\sigma(t) = \lfloor \Sigma(t)/\sigma \rfloor .$$

Equation (E.6) makes Σ nondecreasing, so this ideal calibrated count is nondecreasing and obeys

$$0 \leq \Sigma(t) - \sigma g_\sigma(t) < \sigma, \qquad g_\sigma(t) \leq \left\lfloor \frac{G(0)}{T\sigma} \right\rfloor .$$

The first is the quantization tolerance; the second is the finite energy limit. For $\sigma = 10^{-8}\,\mathrm{J\,K^{-1}}$, the illustrative relaxation can register at most 83 such increments.

A physical implementation measures energy loss and temperature with finite errors. Suppose its energy-loss estimate has error at most ε_D, its temperature estimate error at most ε_T, both true and reported temperatures exceed $T_{\min} > 0$, and the true energy loss is at most $D_{\max}$. Direct subtraction of the two ratios gives

$$\left|\widehat{\Sigma} - \Sigma\right| \leq \frac{\varepsilon_D}{T_{\min}} + \frac{D_{\max}\varepsilon_T}{T_{\min}^2}.$$

This bound must be combined with threshold quantization and sampling delay. A noisy unguarded display can cross the same threshold in both directions. A retained pulse counter instead latches

each accepted crossing; triggering only after a lower error bound passes the next threshold avoids certifying heat that was not established by the measurement. The resulting latency is part of its response specification.

The source's retained construction count has a simpler update law:

$$g_{n+1} = g_n + r_n, \qquad r_n \geq 0.$$

Episode 11 implements successor counting in `universe_id`. It is an accumulating event counter. Its retained count does not by itself establish an increase in every formal universe level; the source notes the maximum-level behavior of `ULift`. Matching physical increments to $\Delta S/\sigma$ requires the calibration above.

A known deterministic symbol can increase record length without increasing Shannon uncertainty. Reversible computation and irreversible erasure also have different physical requirements [2, 16]. Event count, message uncertainty, and thermodynamic entropy therefore retain their separate units and measurement procedures. The beam construction supplies one explicit calibration to dissipated energy, with a finite passive energy supply.

The physical thesis and its controlling estimate

The action closure, scalar digit certificate, and beam dissipation estimate hold under their stated operating assumptions. A stronger statement about fluid singularities requires the relevant material equations and a bound on the physical quantity whose divergence is excluded. Limiting velocity does not by itself limit its spatial gradient, strain rate, vorticity, or the frequency content of a field.

A short exact calculation isolates this distinction. Inviscid Burgers flow obeys $u_t + uu_x = 0$. Use the dimensional initial profile $u_0(x) = -a\sin(kx)$, where a is a speed and k an inverse length. Along the characteristics $x = \xi + tu_0(\xi)$ the speed remains $u_0(\xi)$, but

$$u_x(0,t) = -\frac{ak}{1-akt}.$$

Choosing $0 < a < c$ keeps the speed below c before the first crossing, while its gradient diverges at $t = 1/(ak)$. This calculation concerns an inviscid one-dimensional model. Its purpose is to identify which estimate a finite propagation-speed argument lacks. Viscous fluid regularity must be assessed using its own dissipation, constraints, boundary conditions, and derivative estimates.

A relativistic or microscopic material description can change the validity range of an incompressible continuum approximation. The proposed physical conclusion requires that description, its quantitative fluid limit, and a bound on the relevant stress or gradient. Polynomial degree alone supplies no such bound. The established engineering result is the calibration of an accumulating count to entropy transfer in a beam relaxation whose stationary shape is a variational cubic.

Universe numbers only increase.

The author's following closing sentence states the stronger physical thesis. Its relativistic material model and controlling regularity estimate are additional to the proved constructions in this manuscript.

Finally, this also demonstrates Navier-Stokes cannot blow up in real life because of relativistic phenomena. But, we already knew that.

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